SREYAS COLLEGE ERP

Comprehensive Software Design Document (SDD)

Document Type	Software Design Document (SDD) — Technical Manual
System Architecture	3-Tier Monolithic API Server & SPA Web Portal
Primary Owner	Bhuvanesh T (Project Developer)
Documentation Version	v1.0 (Production Level)
Build Runtime	Node.js v24 · Next.js 16 · Express 4 · pg driver
Database Service	PostgreSQL Serverless Cluster (Neon DB)
Authentication	Stateless JWT Session Handshake & Refresh Cookie
Current Release Ref	Commit 184a34c (Post-MVP Roll list updates)

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CHAPTER 1
Executive Summary

Executive summary outlining the mission, design decisions, and core system architectures of the College ERP.

This Software Design Document (SDD) provides a comprehensive description of the Sreyas College ERP system.

The platform is designed to unify all academic and administrative workflows into a single source of truth, utilizing high-level REST APIs, modular backend services, and clean database relationships.

This document outlines the detailed system design, entity structures, architectural layers, module-specific designs, and verification procedures implemented within the system.

Built on top of a modern Next.js and Express.js framework backed by serverless Neon PostgreSQL database clusters, the system achieves strict role-based access controls, real-time sync networks, and robust transactional integrity.

CHAPTER 2
Project Overview

Detailed overview of the ERP portal operations and integrated system environments.

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1.1 Purpose

The College ERP is a centralized web platform that digitizes the academic and administrative operations of a single college: student/faculty records, attendance, examinations & results, fees, timetabling, learning-management (LMS), mentorship, feedback campaigns, placement/opportunity postings, academic-calendar management, and cross-cutting reporting/analytics. It replaces spreadsheet- and paper-based processes with one relational source of truth, scoped by role-based access control.

1.2 Target Users

Five distinct user populations interact with the system, mapped to 4 database roles plus 2 designation/capability flags layered on top of `faculty`:

Population	DB Role	How they're distinguished
System/College Administrator	`admin`	Role value; an `is_super_admin`…
Head of Department (HOD)	`faculty`	`faculty.designation = 'hod'` (one HO…
Mentoring Head	`faculty`	`faculty.is_mentoring_head = true`…
Regular Faculty	`faculty`	Default designation values (professo…
Student	`student`	Role value; scoped to own records…
Accounts Office Staff	`accountant`	Role value; scoped to Fees module

&**Note**: the planning documents (`project_vis`
Librarian, Parent/Guardian, and Placement Off
`ARCHITECT_REVIEW.md`"Super Admin" as a
of these exist as separate `user_role` enum v
and Librarian/Parent/Placement Officer are n
future-and-appendix.md) for the gap list). The f
layer additionally recognizes a richer set of *RB*
`Academic Coordinator`, `Placement Offi`
nav-filtering purposes — these are **not** backed

the 4 real roles plus designation/flags.

CHAPTER 3
Objectives

Primary goals, target objectives, and scope constraints for the ERP deployment.

1.4 Objectives

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- Maintain an append-only audit trail (`audit_logs`, `workflow_logs`) for security-sensitive actions.

CHAPTER 4
Problem Statement

Analysis of legacy, fragmented, paper-based administrative operations and synchronization gaps.

1.3 Problem Statement

Colleges running manual/spreadsheet-based operations suffer from: duplicated data entry across departments, delayed attendance/grade visibility, no single audit trail for financial or academic record changes, ad hoc scheduling with undetected conflicts, and no unified channel for announcements, assignments, or placement opportunities. The ERP addresses this by making every academic transaction (attendance mark, grade entry, fee payment, teaching assignment) a row in one relational database, visible in real time (via Socket.IO) to every authorized party.

CHAPTER 5
Scope

System functional boundaries, implemented modules, and out-of-scope modules.

1.5 Scope

- **In scope (implemented):** Authentication & RBAC; Student/Faculty/Accountant provisioning; Departments & Programs;

Subjects & Curriculum Mapping; Teaching Assignments; Attendance; Internal Marks; Examinations (scheduling, sessions, seating, invigilation); Results (with a task-

based entry/verify/publish workflow); Fees & Payments; Timetabling; LMS (materials, assignments, submissions/grading); Teaching Plans ("Lesson Progress Engin

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nts; Academic Calendar (including PDF-based auto-extraction of calendar events); Bulk Academic Data Import; Workflow Rule engine; Analytics/Reports/Exports

(PDF/Excel/CSV); Global Search.

- **Out of scope (not implemented, see planning docs' aspirations):** Library management, Hostel/Transport modules, Pare

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1.6 Major Features

See 08-modules.md for the full per-module breakdown. At a glance: 4-role JWT auth with refresh-token rotation and reuse detection · fine-grained + record-level authorization · 43 route files / ~230 REST endpoints · 60 backend tables across 12 functional domains · a task-based Result entry!'verify!'publish approval chain · an auto-balancing mentor-group allocator · a rule-based PDF calendar-event extractor · a generic multi-format (CSV/Excel/PDF) export engine · a configurable Workflow Rule engine driving in-app notifications from domain events · realtime push via Socket.IO for attendance/teaching-plan/exam-seating/ notification updates.

1.7 Future Scope

See 15-future-and-appendix.md §Future Improvements for the full list (containerization/CI, background job queue, Redis caching, payment gateway, AI-driven analytics, Library/ Hostel/Transport/Parent-portal modules, automated testing on the frontend, WebSocket auth hardening).

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CHAPTER 6
Functional Requirements

Detailed breakdown of functional capabilities and operations across portal interfaces.

The functional requirements of the Sreyas College ERP system are structured around the specific operations of individual user portals (Admin, Faculty, Student, and Accountant).

Key functional components include:

- Course & Subject Management: Provisioning of academic records, curriculums, regulations, and allocations.

- Faculty Operations Desk: Real-time attendance logging, grade submission approval workflows, and lesson tracking.

- Student Self-Service: Immediate access to class timetables, attendance status records, examination seat plans, and internal gradebooks.

- Financial Ledger Desk: Custom invoice generation, transaction records tracking, and structure configuration.

CHAPTER 7
Non Functional Requirements

Guarantees on scalability, reliability, transactional safety, and system security.

Non-functional requirements specify constraints on the system's operational behavior. The ERP enforces:

- Scalability: In-process TTL caching (ttlCache) and optimized pg connection pools to handle high concurrent user sessions.

- Security: Stateless JWT verification via httpOnly HTTP cookies, express-rate-limit protection, and record-level filters.

- Reliability: PostgreSQL database-level transactional boundaries (`withTransaction`) enforcing atomic operations.

- Usability: Responsive Tailwind-driven CSS layout structure supporting touch interfaces, mobile navigation grids, and clean grid alignments.

CHAPTER 8
Technology Stack

Comprehensive stack architectural decisions from Next.js 16/React 19 to Neon PostgreSQL serverless.

2. Technology Stack

Layer	Technology	Notes
Frontend framework	**Next.js 16** (App Router), **React…	`next.config.ts` is essentially default…
Frontend styling	**TailwindCSS 4** (+ `@tailwindcss/…	Utility-first; dark/light theme via CSS…
Frontend data/state	**@tanstack/react-query 5** (partial…	No Redux/Zustand; see [04-…
Frontend charts	**Recharts 3**	Dashboard/analytics visualizations
Frontend realtime	**socket.io-client 4.8**	Auth via `{token: accessToken}`…
Icons	**lucide-react**	
Backend runtime	**Node.js**, **Express.js 4**,…	No web framework beyond Express;…
Backend DB access	**pg 8** (raw SQL, hand-written…	See 05-backend.md
Database	**PostgreSQL** (Neon serverless…	50 versioned SQL migrations,…
Auth	**jsonwebtoken 9** (HS256, separat…	See [07-authentication-and-…
Validation	**Zod 3** (both a request-validation…	
Realtime (server)	**socket.io 4.8**	In-process `eventBus.service.ts` pub…
File processing	**multer 2** (uploads), **exceljs 4**…	
Security middleware	**helmet 8**, **cors 2.8**, **express-…	
Testing	**Jest 29** + **ts-jest** (backend onl…	
Build tools	`tsc` (backend compile to `dist/`),…	
Package manager	npm (both `frontend/` and `backend/`…	
Dev tools	`tsx` (backend dev/hot-reload runner…	
Version control	Git (this repository)	No CI/CD pipeline configured (no…

Full dependency-by-dependency breakdown
deployment-config-dependencies.md) §Depen

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CHAPTER 9
High Level Architecture

High-level tier, interface, and structural diagram mappings.

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CHAPTER 10
System Architecture

Layered architecture pattern mapping controllers, services, and transactions.

2.1 Layered System Architecture

The system is a classic 3-tier web application: a Next.js client, a monolithic Express API, and a PostgreSQL database — plus a realtime side-channel (Socket.IO) and a handful of external delegation points (email/SMTP is referenced in planning docs but **not implemented**; there is no external payment gateway; PDF/Excel generation happens in-process).

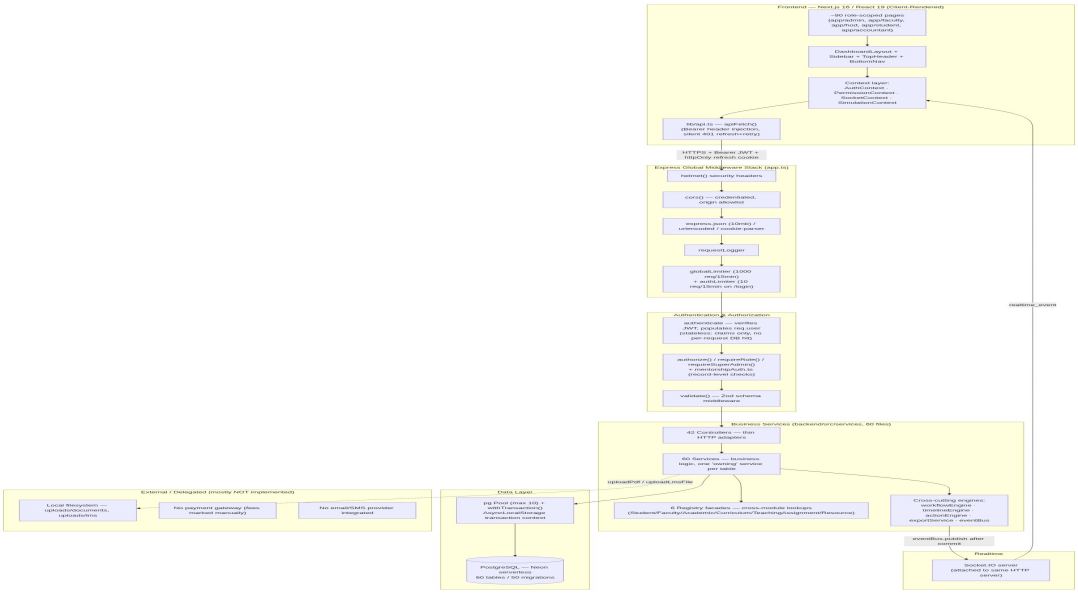


Figure 1: Diagram mapping for 2.1 Layered System Architecture

Layer responsibilities

Layer	Responsibility	Key files
Frontend (Client)	Renders role-scoped UI, holds…	`frontend/src/app/**`, `frontend/src/li…
API Layer (Express routers)	Declares URL surface, wires…	`backend/src/routes/*.ts` (43 files)
Authentication	Verifies JWT, issues/rotates tokens,…	`middleware/authenticate.ts`, `utils/…
Authorization	Role/permission/record-level gating	`middleware/authorize.ts`,…
Business Services	Domain logic, one owning service pe…	`backend/src/services/*.ts` (60 files),…
Database	Persistent relational storage, enforce…	`backend/src/db/migrations/*.sql` (50…
External Services	File storage (local disk only today); n…	`middleware/upload.ts`, `middleware/…

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2.2 High-Level Role Interaction Architecture

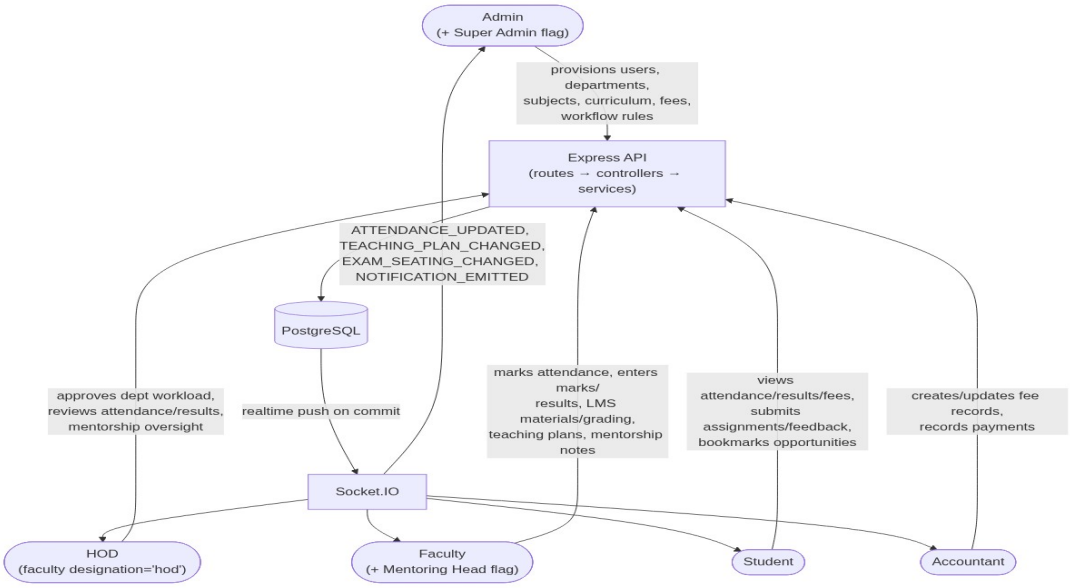


Figure 2: Diagram mapping for 2.2 High-Level Role Interaction Architecture

Notes on the diagram

- HOD and Mentoring Head

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`isSuperAdmin`/`facultyId`/`departmentId`/`studentId` claims are what differentiate downstream authorization (see [07-authentication-and-security.md](07-authenti

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2.3 Frontend Layout Composition (per-role)

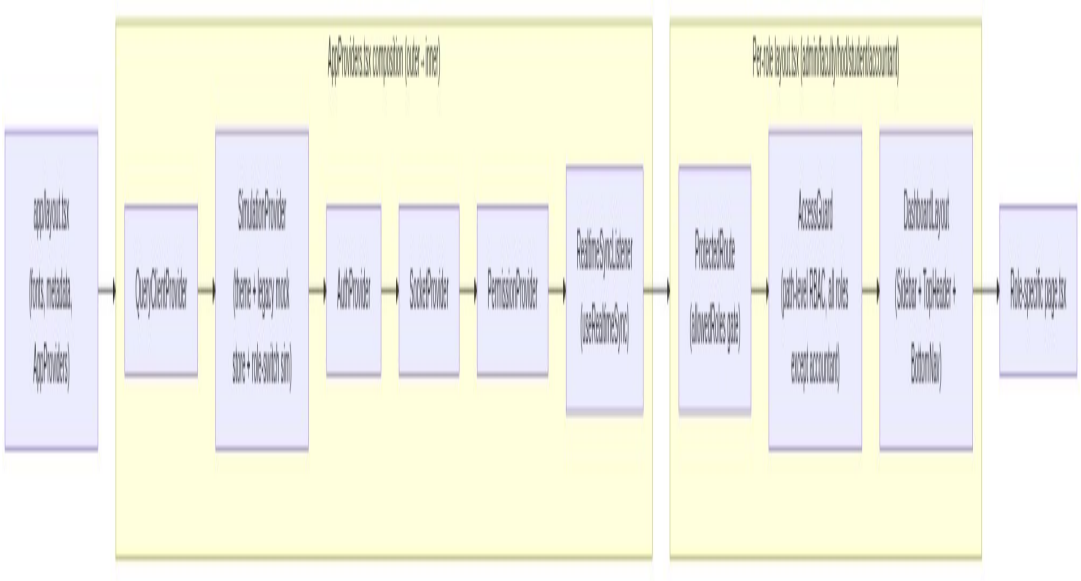


Figure 3: Diagram mapping for 2.3 Frontend Layout Composition (per-role)

See 04-frontend.md for full detail on each box.

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CHAPTER 11
Frontend Architecture

Next.js App Router workspace patterns, authentication context, and client-side guards.

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The frontend is a Next.js 16 (App Router) application, effectively operating as a **client-rendered SPA**that calls an external Express API — `next.config.ts`is default/empty, there are no server actions, API routes, or SSR data-fetching patterns in use; every page is a `"use client"`component fetching data imperatively after mount.

4.1 Root & Role Layouts

`frontend/src/app/layout.tsx` (server component) sets up `Inter`/`Outfit`Google Fonts as CSS variables, global metadata ("Sreyas College ERP - Academic Portal"), imports `globals.css`, and wraps `{children}` in the single `<AppProviders>` composition root.

Each role subtree has its own `layout.tsx`, all client components with an almost identical shape:

```
ProtectedRoute (role gate) !’ AccessGuard (path-level RBAC) !’ DashboardLayout (chrome) !’
{children}
```

Layout	`allowedRoles`	Extra logic
`app/admin/layout.tsx`	`["admin"]`	—
`app/faculty/layout.tsx`	`["faculty"]`	—
`app/hod/layout.tsx`	`["faculty"]`	`useEffect` double-checks…
`app/student/layout.tsx`	`["student"]`	—
`app/accountant/layout.tsx`	`["accountant"]`	**Does not wrap `AccessGuard`** —…

&**Note**: No `app/loading.tsx`, `app/error.t`
exist anywhere in the tree — Next.js route-leve
is handled ad hoc inside individual component

4.12 Frontend Architecture Summary

The frontend intentionally has **no backend-for-frontend layer**— Next.js's own API-route capability is unused; it is a pure client calling the Express API directly via `NEXT_PUBLIC_API_URL`. This keeps the two codebases fully decoupled (any client could replace the Next.js frontend without backend changes) at the cost of duplicating some type definitions between `frontend/src/lib/*.ts` / `frontend/src/types/*.ts` and `backend/src/types/*.ts`(no shared/generated types package exists — see 15-future-and-appendix.md for this as a noted improvement opportunity).

CHAPTER 12
Backend Architecture

Node/Express route definitions, middleware stack, and business service layers.

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The backend is a monolithic Express 4 + TypeScript application (CommonJS, compiled via `tsc`), with **no ORM** — every query is hand-written SQL against the `pg` driver. It follows a strict **Route !’ Controller !’ Service !’ Database** layering, plus a **Registry** facade layer for cross-module reads (see 02-architecture-and-diagrams.md).

5.1 Application Entry & Global Middleware

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exit.

- **`app.ts`** (`createApp()`) — global middleware stack, in order:

1. `app.set('trust proxy', 1)` — correct `req.ip` behind a proxy (used by

rate limiter and audit logging).

2. `helmet()` — security headers.
3. `cors({ origin: env.CORS_ORIGIN.split(','), credentials: true, methods: GET/POST/PUT/PAT

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6. `requestLogger` — logs `METHOD URL STATUS DURATIONms`.
7. `globalLimiter` — 1000 req/15min per IP.
8. `app.use('/api', apiRouter)` — mounts

the entire route tree.

9. 404 handler.
10. `errorHandler` — last middleware (§5.5).

A stricter `authLimiter` (10 req/15min) applies only to `POST /api/auth/login`.

5.2 Routes (43 files, ~230 endpoints)

Every route file mounts under `/api` via `routes/index.ts`. Full endpoint-by-endpoint reference: **09-api-documentation.md**. Summary table:

Base path	File	Domain
`/auth`	auth.routes.ts	Login/refresh/logout/me
`/search`	search.routes.ts	Global search
`/profile`	profile.routes.ts	Own-account profile/password
`/dashboard`	dashboard.routes.ts	Per-role dashboard aggregation
`/accountants`	accountant.routes.ts	Accountant account CRUD
`/departments`	departments.routes.ts	Departments & Programs
`/students`	students.routes.ts	Student CRUD & listing
`/faculty`, `/faculty-operations`	faculty.routes.ts,…	Faculty CRUD; faculty operations…
`/subjects`	subjects.routes.ts	Subject catalog, curriculum mapping…
`/attendance`	attendance.routes.ts	Attendance assignments, roster,…
`/internal-marks`	internal-marks.routes.ts	Internal assessment marks
`/examinations`, `/exam-sessions`, `/…	(6 files)	Examination scheduling/seating/…
`/results`, `/result-entry-tasks`	result.routes.ts,…	Results + task-based entry/verify/…
`/fees`	fee.routes.ts	Fee invoices & payments
`/announcements`	announcement.routes.ts	Institutional announcements
`/documents`, `/parsed-events`, `/…	document.routes.ts,…	PDF upload !' rule-based event…
`/lms/materials`, `/lms/assignments`,…	material/lmsAssignment/…	LMS
`/opportunities`	opportunityHub.routes.ts	Placement/internship/event postings…
`/notifications`, `/calendar-entries`	notification.routes.ts,…	Notifications; personal calendar feed
`/mentorship`, `/mentor-groups`, `/…	(3 files)	Mentorship (legacy 1:1 + group-based)
`/analytics`, `/reports`, `/experience`	(3 files)	Analytics, reports/exports,…
`/feedback`	feedback.routes.ts	Feedback campaigns
`/workflow`	workflowRule.routes.ts	Configurable notification/automation…
`/subject-allocations` (deprecated), `/…	(4 files)	Teaching workload assignment &…
`/academic-import`	academicImport.routes.ts	Generic bulk spreadsheet import…
`/timetables`	timetable.routes.ts	Timetable generation/publishing
`/example`	protected.routes.ts	Reference implementation of the 6…
`/health`	inline in `app.ts`	Liveness check, no auth

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5.3 Controllers

42 controller files, one roughly per route file — thin HTTP adapters: parse/validate is already done by middleware, so a controller typically extracts `req.user`/`req.params`/`req.body`, calls exactly one service method, and returns via `sendSuccess`/`sendCreated` (or throws an `AppError`, caught by the global handler). No business logic lives here.

5.4 Services (60 files) — Ownership Model

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.md) for the full per-module service breakdown

and 09-api-documentation.md for which tables each service touches.

- **Cross-cutting / infrastructure services (no direct routes):**

Service	Purpose
`eventBus.service.ts`	Singleton wrapping the Socket.IO server; `publish(type,…
`export.service.ts`	Shared Excel/PDF/CSV renderer (ExcelJS + PDFKit), 50…
`actionEngine.service.ts` / `timelineEngine.service.ts`	Student/Faculty "Experience" aggregation-and-rules layer…
`workflowEngine.service.ts`	Generic rule execution + audit engine (`workflow_rules`,…
`facultyDailyStatus.service.ts`, `facultyTasks.service.ts`,…	Faculty Operations Center support aggregations
`dashboardAggregator.service.ts`	Shared aggregation helper underlying per-role dashboard…
`resourceAvailability.service.ts`	Shared room/invigilator availability engine used by exam-…

5.5 Registries (`backend/src/registries/`, 6 files + barrel)

An internal, non-routed "master data" facade layer — no independent tables of their own. Purpose: prevent the ~20-45 ad hoc duplicate lookup queries that existed before consolidation.

Registry	Wraps	Consolidates
`studentRegistry.service.ts`	`student.service.ts`	Core student-identity lookups (roll no…
`facultyRegistry.service.ts`	`faculty.service.ts`	Core faculty-identity lookups —…
`academicRegistry.service.ts`	`department.service.ts`	Departments/Programs/"Academic…
`curriculumRegistry.service.ts`	`subject_curriculum_mappings`	Resolves "what should this student/…
`teachingAssignmentRegistry.servic…	`teachingAssignment.service.ts` +…	The shared "is this faculty assigned t…
`resourceRegistry.service.ts`	`examRoom.service.ts`	"What rooms exist / are available" for…

5.6 Middleware (`backend/src/middleware/`)

Full detail in 07-authentication-and-security.md. Summary:

File	Purpose
`authenticate.ts`	Bearer/`?token=` JWT verification !' populates `req.user`
`authorize.ts`	`authorize()` (permission matrix), `requireRole()` (coarse…
`mentorshipAuth.ts`	Record-level authorization for Mentorship (mentor/mentee…
`rateLimiter.ts`	`globalLimiter` (1000/15min), `authLimiter` (10/15min on…
`errorHandler.ts`	Global error handler (§5.9)
`requestLogger.ts`	Console request/response logging
`upload.ts`	`uploadPdf` — disk storage, 20MB, PDF-only (Documents)
`uploadLms.ts`	`uploadLmsFile` — disk storage, 20MB, PDF/PPT/PPTX/…
`validate.ts`	Zod schema validation middleware factory for body/param…

Two additional ad hoc multer instances exist inline in route files 10MB in `subjects.routes.ts` (Excel/CSV import preview) an `academicImport.routes.ts`(generic spreadsheet import) — disk.

CHAPTER 13
Database Architecture

Normalization strategy, schema constraints, indexes, and transactional boundaries.

!• [Back to Index](00-INDEX.md)

- **Engine**: PostgreSQL (Neon serverless in practice).
Schema history: 50 sequential numbered migrations (`backend/src/db/

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6.1 Global Conventions

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6.2 Enum Types

Enum	Introduced	Final value set
`user_role`	001 (+021)	admin, faculty, student, accountant
`student_status`	002	active, graduated, suspended, inactive
`faculty_designation`	003 (+020)	professor, associate_professor,…
`faculty_status`	003	active, on_leave, resigned, retired
`subject_type`	004 (+044)	core, elective, lab, project, seminar *…
`subject_status`	004	active, inactive, archived
`attendance_status`	006	present, absent
`assessment_type`	007	Assignment, Mid-1, Mid-2, Lab,…
`exam_type`	008 (+039)	Mid-1, Mid-2, Lab Exam, Internal, En…
`exam_status`	008 (+045)	Scheduled, Ongoing, Completed,…
`result_grade`	009	O, A+, A, B+, B, C, F
`result_status`	009	Pass, Fail, Absent
`result_pub_status`	009	Draft, Published
`fee_type`	010	Tuition Fee, Examination Fee,…
`payment_status`	010	Pending, Partially Paid, Paid, Overdue
`fee_payment_mode`	010	Cash, DD, Cheque, Online
`announcement_audience`	011	All, Students, Faculty, Admin,…
`announcement_priority`	011	Low, Medium, High, Urgent
`announcement_status`	011	Draft, Published, Expired
`document_type`	013	Academic Calendar
`document_status`	013	Uploaded, Processed
`parsed_event_type`	014	Class Commencement, Mid-Term…
`parsed_event_audience`	014	All, Students, Faculty, I/II/III/IV Year
`parsed_event_status`	014	Pending, Approved, Edited, Rejected
`cal_publish_status`	015	Published, Updated, Archived
`lms_file_type`	017	pdf, ppt, pptx, doc, docx
`submission_status`	017	Submitted, Late Submission,…
`opportunity_type`	018	Internship, Job Opportunity,…
`opportunity_status`	018	Active, Closed, Archived
`notification_type`	019	Announcement, Assignment, Grade…
`calendar_event_type`	019	Academic, Assignment Deadline,…
`calendar_visibility`	019	personal, department, semester,…
`mentor_assignment_status`	023	active, reassigned, completed
`mentor_assignment_method`	024	range, section, manual
`lesson_completion_status`	025 (+026)	Planned, Completed, Rescheduled,…

- **Fixed value-sets not backed by an enum** (VARCHAR + CHECK): `teaching_assignments.status` ('active'/'inactive'/'pending'), `examination_sessions.status` (V

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6.3 Entity-Relationship Diagrams

Given 60 tables, the ER model is presented as several domain-scoped diagrams rather than one unreadable diagram.

6.3.1 Core Identity & Organization

```
erDiagram
    USERS ||--o| STUDENTS : "user_id"
    USERS ||--o| FACULTY : "user_id"
    USERS ||--o| ACCOUNTANTS : "user_id"
    USERS ||--o{ REFRESH_TOKENS : "user_id"
    USERS ||--o{ AUDIT_LOGS : "actor_id"
    DEPARTMENTS ||--o{ PROGRAMS : "department_id"
    DEPARTMENTS ||--o{ STUDENTS : "department_id"
    DEPARTMENTS ||--o{ FACULTY : "department_id"
    PROGRAMS ||--o{ STUDENTS : "program_id"

    USERS { uuid id PK }
    STUDENTS { uuid id PK, uuid user_id FK, varchar roll_number UK, uuid department_id FK,
uuid program_id FK, smallint semester, varchar academic_year }
    FACULTY { uuid id PK, uuid user_id FK, varchar employee_number UK, uuid department_id FK,
enum designation, bool is_mentoring_head }
    ACCOUNTANTS { uuid id PK, uuid user_id FK, varchar employee_number UK }
    DEPARTMENTS { uuid id PK, varchar name UK, varchar code UK }
    PROGRAMS { uuid id PK, uuid department_id FK, varchar code UK }
```

`faculty` additionally enforces **one HOD per department** via a partial unique index: `idx_unique_hod_per_dept ON faculty(department_id) WHERE designation='hod' AND deleted_at IS NULL`.

CHAPTER 14
Folder Structure

Full path structures and file responsibilities within the workspace.

!• [Back to Index](00-INDEX.md)

3.1 Repository Root

```
ERP/
% % %  frontend/                    # Next.js 16 client application
% % %  backend/                     # Express/TypeScript API server
% % %  docs/                        # Miscellaneous engineering docs + this architecture document
%    % % %  architecture/            # !• this Software Architecture Document
% % %  AGENTS.md                    # AI-agent operating instructions for this repo
% % %  CLAUDE.md                    # Points to AGENTS.md
% % %  README.md                    # Project intro, setup instructions
% % %  project_vision_and_requirements.md   # Original planning doc (aspirational, MVP-oriented)
% % %  ARCHITECT_REVIEW.md          # Early architecture review notes
% % %  SYSTEM_MODEL.md              # Conceptual entity/actor model (planning-stage)
% % %  BUSINESS_RULES_AND_WORKFLOWS.md      # Planning-stage business rules
% % %  MVP_DEFINITION_AND_DEVELOPMENT_ROADMAP.md   # Original 3-week MVP roadmap
% % %  UI_UX_BLUEPRINT.md           # UI/UX design system spec (mostly matches implementation)
% % %  walkthrough.md               # Dev log for the Subjects Master module
% % %  sample_student_import.csv    # Sample file for the student bulk-import feature
% % %  I SEMESTER ROLL LIST WITH FULL DATA.xlsx   # Real sample data used with the import pipeline
```

&**Note**: the planning `.md`files at the root describe an earlier, smaller-scope MVP (3 roles, no approval workflows). They are retained for historical context but should not be treated as current-state documentation — this architecture document (`docs/architecture/`) supersedes them for "what the system actually does."

- ---

3.2 Backend — `backend/src/`

```
backend/src/
% % %  server.ts              # Process entrypoint: pg connectivity check, HTTP+Socket.IO server,
graceful shutdown
% % %  app.ts                 # createApp(): global Express middleware stack + route mounting (see
02-architecture)
% % %  config/
%    % % %  env.ts              # Zod-validated environment schema; fail-fast on invalid config
%    % % %  database.ts         # pg Pool, query() helper, withTransaction() + AsyncLocalStorage
event context
%    % % %  grading.ts          # Grade-boundary constants (O/A+/A/B+/B/C/F) + computeGrade/
computeResultStatus
% % %  routes/                 # 43 files — one per resource, thin: middleware chain + controller
wiring only
% % %  controllers/            # 42 files — HTTP adapters: parse req, call service, sendSuccess/
sendError
% % %  services/               # 60 files — business logic; one "owning" service per table (see 05-
backend.md)
% % %  registries/             # 6 facade services — cross-module lookups (Student/Faculty/
Academic/Curriculum/TeachingAssignment/Resource) + index.ts barrel
% % %  middleware/
%    % % %  authenticate.ts       # JWT verification ’ req.user
%    % % %  authorize.ts          # authorize()/requireRole()/requireSuperAdmin() — static RBAC
%    % % %  mentorshipAuth.ts     # Record-level (mentor/mentee/note-owner) authorization
%    % % %  validate.ts           # Zod schema validation middleware factory
%    % % %  rateLimiter.ts        # globalLimiter + authLimiter (express-rate-limit)
%    % % %  errorHandler.ts       # Global error handler (Postgres error code mapping + AppError
formatting)
%    % % %  requestLogger.ts      # Console-based request/response logging
%    % % %  upload.ts             # Multer config — generic PDF documents (disk storage)
%    % % %  uploadLms.ts          # Multer config — LMS materials (PDF/PPT/DOC, disk storage)
% % %  errors/
%    % % %  AppError.ts           # Single error class + static factories (badRequest/unauthorized/
forbidden/notFound/conflict/internal)
% % %  import/                 # Generic entity-agnostic bulk-import pipeline framework
%    % % %  ExcelParser.ts         # .xlsx/.xls/.csv ’ row-object parser (header auto-detection)
%    % % %  ImportStrategy.ts      # Interface every entity importer must implement
%    % % %  ImportTransactionManager.ts   # Chunked, single-transaction commit orchestration
%    % % %  ValidationPipeline.ts  # parse ’ validate ’ summarize glue
%    % % %  strategyRegistry.ts    # ImportEntityType ’ strategy implementation map
%    % % %  strategies/
%        % % %  studentImport.strategy.ts   # The only implemented strategy (Merge/Replace modes)
% % %  db/
%    % % %  migrate.ts             # Migration runner (handles enum ALTER TYPE special-casing)
%    % % %  seed.ts, seed_exam_seating.ts, seed_feedback.ts   # Seed data scripts
%    % % %  check.ts               # DB connectivity/sanity check script
%    % % %  sync_roll_list.ts      # One-off script for syncing the Excel roll-list sample data
%    % % %  query_student.ts, query_opportunities.ts, list_users_test.ts   # Ad hoc query/debug
scripts
%    % % %  migrations/            # 50 numbered .sql files (001_... through
050_timetable_management.sql)
% % %  types/                  # Shared TypeScript types + Zod schemas per domain (roles.ts,
auth.ts, subject.ts, etc.)
% % %  utils/
%    % % %  asyncHandler.ts        # Wraps async route handlers ’ forwards rejections to next()
%    % % %  audit.ts                # auditLog() ’ audit_logs table
%    % % %  calendarExtractor.ts    # Rule-based PDF-text ’ academic calendar event extractor
%    % % %  jwt.ts                  # Access/refresh token sign/verify
%    % % %  password.ts             # bcryptjs hash/verify
```

```
%    % % %  response.ts             # sendSuccess/sendCreated/sendError envelope helpers
%    % % %  ttlCache.ts             # In-process Map-based TTL memoizer (no Redis)
%    % % %  validation.ts           # Password strength regex + Zod schema
% % %  __tests__/                # 20 Jest test files (service-level, ts-jest)

backend/uploads/
% % %  documents/                # uploadPdf target (Document Management)
% % %  lms/                      # uploadLmsFile target (Course Materials, Submissions)
```

- ---

3.3 Frontend — `frontend/src/`

```
frontend/src/
% % %  app/                      # Next.js App Router — file-based routing
%    % % %  layout.tsx             # Root layout: fonts, metadata, <AppProviders>
%    % % %  page.tsx               # Login page (root "/")
%    % % %  admin/                # 27 routes — layout.tsx gates role="admin"
%    %    % % %  layout.tsx
%    %    % % %  accountants/ analytics/ announcements/ attendance/ calendar/ courses/
%    %    % % %  dashboard/ examinations/ faculty/ feedback/ fees/ import/ lms/[subjectId]/
%    %    % % %  mentorship/ notifications/ opportunities/ profile/ reports/ results/
%    %    % % %  seating/ settings/ students/ subject-allocations/ subjects/[id]/
%    %    % % %  teaching-planner/ timetables/ workflows/
%    % % %  faculty/                # 18 routes — layout.tsx gates role="faculty"
%    %    % % %  layout.tsx
%    %    % % %  analytics/ announcements/ attendance/ calendar/ dashboard/ examinations/
%    %    % % %  feedback/ grades/ invigilation/ lms/[subjectId]/
%    %    % % %  mentorship/[groupId]/[studentId]/ notifications/ opportunities/ profile/
%    %    % % %  reports/ results/ subjects/ teaching-planner/ timetable/ workflows/
%    % % %  hod/                    # 12 routes — layout.tsx gates role="faculty" +
designation==="hod" check
%    %    % % %  layout.tsx
%    %    % % %  analytics/ attendance/ classes/ dashboard/ faculty/ feedback/
%    %         mentorship/ reports/ seating/ students/ subject-allocations/ subjects/
%    %         (HOD reuses several /faculty/* routes directly for teaching-workload duties)
%    % % %  student/                # 17 routes — layout.tsx gates role="student"
%    %    % % %  layout.tsx
%    %    % % %  analytics/ announcements/ attendance/ calendar/ dashboard/ examinations/
%    %         feedback/submit/ fees/ lms/[subjectId]/ mentorship/ notifications/
%    %         opportunities/[id]/bookmarks/ profile/ reports/ results/ seating/ timetable/
%    % % %  accountant/             # 2 routes — layout.tsx gates role="accountant" (no AccessGuard
layer)
%    %    % % %  layout.tsx
%    %    % % %  dashboard/ fees/
%    % % %  unauthorized/           # Fallback page for AccessGuard/PermissionContext denial
% % %  components/               # Shared UI (see 04-frontend.md §Component Architecture for full
file list)
%    % % %  DashboardLayout.tsx, Sidebar.tsx, TopHeader.tsx, BottomNav.tsx, ProtectedRoute.tsx
%    % % %  CalendarView.tsx, EventFormModal.tsx, PersonalEventModal.tsx, NotificationCenter.tsx
%    % % %  FacultyFormDrawer.tsx, StudentFormDrawer.tsx, DeleteConfirmationModal.tsx,
FacultyDeleteModal.tsx
%    % % %  OpportunityCard.tsx, OpportunityFilterBar.tsx, OpportunityFormModal.tsx,
PDFUploadModal.tsx
%    % % %  Analytics/    # AccessGuard, AcademicSubNav, AnalyticsCharts, InsightCards, FilterPanel,
ExportDialog, ProtectedComponentWrapper
%    % % %  Dashboard/    # StatsCard, ActivityFeed, AssignmentWidget, CalendarWidget,
DashboardCharts, OpportunityWidget, UpcomingEventsWidget, NotificationWidget,
TeachingProgressCard, TodayTaskCenter, WorkloadCharts, MyTeachingAssignmentsCard
%    % % %  Feedback/     # AnonymousCommentList, FeedbackCharts, CampaignForm, CampaignManager,
FeedbackTemplatesManager, FeedbackWizard, SubjectFeedbackCard
%    % % %  LMS/          # LMSDashboard, SubjectLMS, Modals/
{AssignmentModal,GradingModal,MaterialModal,SubmissionModal}
%    % % %  Profile/      # ProfileOverview, EditProfileForm, ChangePasswordForm
%    % % %  Search/       # GlobalSearch
%    % % %  Seating/      # SeatingChartGrid, PatternFormModal, RoomFormModal, SessionFormModal
%    % % %  SubjectAllocation/   # AllocationManager
%    % % %  Subjects/     # SubjectCatalogManager
%    % % %  TeachingAssignment/  # TeachingAssignmentManager
%    % % %  UX/           # ConfirmationDialog, EmptyState, SkeletonLoaders, UnifiedPagination,
UnifiedSearchBar
```

```
%    % % %  UXL/          # ActionCenter, QuickActions, SmartProgressCard, TodayFocus, UnifiedTimeline
% % %  context/
%    % % %  AuthContext.tsx         # Real backend session (user/token/login/logout/refresh)
%    % % %  SimulationContext.tsx   # Legacy/demo local-state store + global theme + role-switch
simulator (localStorage-persisted)
%    % % %  SocketContext.tsx       # Socket.IO client connection
%    % % %  PermissionContext.tsx   # Derived RBAC role + path/widget access rules
% % %  providers/
%    % % %  AppProviders.tsx        # Composition root (QueryClient !’ Simulation !’ Auth !’ Socket !’
Permission !’ RealtimeSyncListener)
% % %  hooks/
%    % % %  useRealtimeSync.ts      # Socket event !’ React Query cache invalidation + custom DOM events
%    % % %  useRegistry.ts          # React Query hooks wrapping apiFetch for shared "registry"
lookups
% % %  lib/
%    % % %  api.ts                  # apiFetch() — auth header injection, silent 401 refresh+retry
%    % % %  permissions.ts          # RBAC role resolution + path/widget access rule tables
%    % % %  lms.ts, opportunities.ts, search.ts, profile.ts, seating.ts, teachingPlan.ts,
feedback.ts, mentorship.ts   # Domain type definitions
% % %  types/
%    % % %  opportunity.ts, subject.ts, academicImport.ts, realtime.ts, subjectAllocation.ts
% % %  globals.css               # Tailwind base + design tokens
```

CHAPTER 15
Authentication Flow

JWT-based session authentication with refresh-token rotation and cookie storage rules.

4.2 Authentication Flow

- **Login page** (`frontend/src/app/page.tsx`, root `/`): a single controlled-input form calling `useAuth().login(username, passw

ord)
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_has_session = "true"` used to attempt silent refresh on load.

- **Session bootstrap**: on mount, if the hint is set, calls `POST /auth/refresh` with `credential

s: "include"` (relying on the http Only refresh cookie); on success, populates token/user and syncs into `SimulationContext`.

- Listens for two custom `window` ev

ents dispatched by `lib/api.ts`: `tokenRefreshed` (keeps context in sync after a silent refresh triggered by a 401) and `sessionExpired` (clears state when refresh

fails
).

- `logout()` calls `POST /auth/logout`, clears state + the localStorage hint, and hard-navigates to `/` via `window.location.href`.
- **`lib/api.ts`** (`apiFetch<

T>(endpoint, options, accessToken?)`):

- Attaches `Authorization: Bearer <token>` (explicit arg or module-level `globalAccessToken` fallback); auto-sets `Co

nten t-Ty pe: appl icati on/j son` unl ess the bod y is `For mD ata` .

- **S ilent refr esh- and -retr y**: on a 401 (exc ept for `/au th/r efre sh`/ `/au th/lo gin` the mse lves) it c alls

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- Network failures return a soft `{success:false, message:"Backend server is currently offline..."}` object instead of throwing, so callers can render inline er

rors rath er t han cras h.

- `A PI_ URL ` de fault s to `htt p://1 27.0 .0.1: 300 1/ api` via ` NE XT_ PU BLI C_A PI_ URL `.
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4.3 Role Protection (3 layers, defense in depth)

1. **`components/ProtectedRoute.tsx`** — coarse role gate. Shows a full-screen "Verifying Identity Session..." spinner while `loading`; redirects to `/` if no `

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*`lib/permissions.ts`** — the RBAC engine:

- `getRbacRole()` resolves a fine-grained label (`Super Admin`, `College Admin`, `HOD`, `Academic Coordinator`,

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> **Note**: no root-level `middleware.ts` exists — route protection is entirely client-side/component-based, not edge middleware. A determined client could briefly see a flash of protected route structure before the client-side redirect fires (standard SPA-auth caveat).

4.4 Page Routes by Role

Full path listing is in 03-folder-structure.md. Route counts: **Admin 27**, **Faculty 18**, **HOD 12** (+ reuses several `/faculty/*` routes for teaching-workload duties per `Sidebar.tsx`'s "Teaching Desk" nav group), **Student 17**, **Accountant 2**. Total "H 76 distinct top-level page routes plus dynamic segments (`[subjectId]`, `[id]`, `[groupId]`, `[groupId]/[studentId]`).

Representative purpose per role (full table in 03-folder-structure.md):

- **Admin**: system-wide CRUD registries (students/faculty/accountants/subjects/departments via courses), curriculum configuration, teaching-assignment allo

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- **HOD**: department-scoped oversight dashboards (workload charts, teaching-progress cards, action center) plus the shared faculty teaching-workspace rout

es.

- **Student**: read-mostly consumption views (attendance, results, fees, timetable, LMS, seating) plus write actions limited to assignment submissions, feedback,

and opportunity bookmarks.

- **Accountant**: fee ledger and dashboard only.

4.5 Shared Component Architecture

- **Top-level `components/`** (cross-role chrome): `DashboardLa

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.tsx` (calendar building blocks reused across 4 roles), `NotificationCenter.tsx` (full inbox: tabs, filters, pagination, mark read/unread), `FacultyFormDrawer.tsx`/`St

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- **D oma in s ubfo lder s** (full listin g in [03- fold er-st ruct ure. md] (03- fold er-st ruct ure. md)): `A naly tics/ ` (R BA C wi dget

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4.6 Hooks, Lib, Providers, Context, Types

Folder/file	Purpose
`hooks/useRealtimeSync.ts`	Subscribes to the Socket.IO `realtime_event` channel;…
`hooks/useRegistry.ts`	Shared React Query hooks wrapping `apiFetch`:…
`lib/api.ts`	Core fetch wrapper (§4.2)
`lib/permissions.ts`	RBAC engine (§4.3)
`lib/lms.ts`, `lib/opportunities.ts`, `lib/search.ts`, `lib/…	Per-domain TypeScript type definitions (no runtime logic…
`providers/AppProviders.tsx`	Composition root: `QueryClientProvider` (staleTime 60s,…
`context/AuthContext.tsx`	Real backend session (§4.2)
`context/SimulationContext.tsx`	**Legacy/demo local-state store**, `localStorage`-persiste…
`context/SocketContext.tsx`	Socket.IO client connection (§4.2)
`context/PermissionContext.tsx`	RBAC engine wiring (§4.3)
`types/*.ts`	`opportunity.ts`, `subject.ts`, `academicImport.ts`,…

4.7 State Management

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5.7 Authentication & Authorization

See 07-authentication-and-security.md for the complete mechanics (JWT signing, refresh rotation with reuse detection, bcrypt cost factor, permission matrix, record-level checks).

!• [Back to Index](00-INDEX.md)

7.1 Login Flow

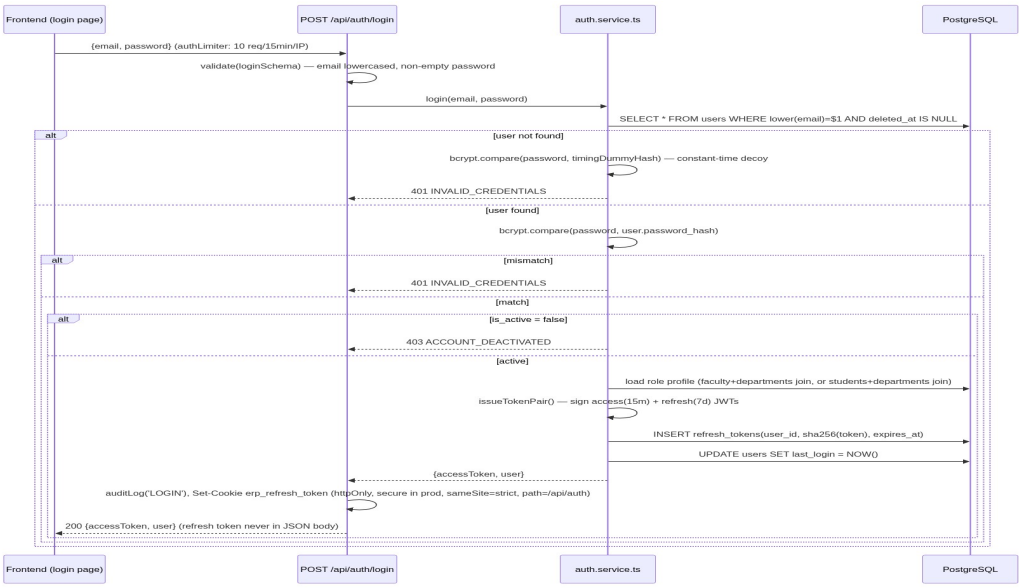


Figure 5: Diagram mapping for 7.1 Login Flow

- **Timing-attack mitigation**: when the email doesn't exi

st, the service still performs a `bcrypt.compare` against a lazily-computed dummy hash, so the response latency for "no such user" and "wrong password" is in

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7.2 JWT Design

Property	Access Token	Refresh Token
Secret	`JWT_ACCESS_SECRET` (min 32…	`JWT_REFRESH_SECRET`…
TTL (default)	15 minutes	7 days
Algorithm	HS256 (implicit — plain string secret…	HS256
Payload claims	`sub` (user id), `email`, `role`,…	`sub`, `tokenId` (uuid, currently…
Delivered via	JSON response body	httpOnly cookie only…
Storage on server	Not stored (stateless verification)	`sha256(token)` hash stored in…
Verification	`authenticate` middleware on every…	Only checked at `/auth/refresh`

&**Design implication**: because the access toke
`authenticate.ts` trusts those claims for up
deactivation, or HOD reassignment made by
token until it expires or the user re-authenticate
it always re-fetches from the DB "so the respon

7.3 Refresh Token Rotation & Reuse Detection

On `POST /api/auth/refresh`:

1. Verify the refresh JWT's signature/expiry (fails fa

st, no DB hit if invalid).

2. Hash the token, look up `refresh_tokens` joined to `users` by `(token_hash, user_id)`.
3. Not found !’ ` 401 TOKEN_INVALID`.

4. `expires_at` in the past !’ `401 TOKEN_EXPIRED`.
5. **Reuse detection**: if `revoked_at IS NOT NULL` (this exact token was already consumed once) !’ tre

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`logout()` revokes just the one token matching the presented cookie. `logout-all` (`POST /api/auth/logout-all`, requires a valid access token) revokes every active token for the current user — used after password changes (`profile.service.ts`'s `changePassword` calls this implicitly by design intent, invalidating all other sessions).

Gap: no scheduled job purges expired/revoked `refresh_tokens` rows — the table grows unbounded over the system's lifetime (see [12](12-performance-error-handling-workflow.md) and [15](15-future-and-appendix.md)).

7.4 Password Handling

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ure-JS, not native `bcrypt`), cost factor from `env.BCRYPT_ROUNDS` (Zod-bounded 10–14, default **12**).

- **Strength policy** (`utils/validation.ts`, enforced at

set-time — registration/reset/change-password, **not** at login): minimum 8 characters, must include upper, lower, digit, and one of `@$!%*?&`.

- **Change password**

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8.1 Identity & Access (Authentication, Users, Roles)

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- **Workflow**: Login → JWT issuance → per-request stateless verification → refresh rotation → logout/logout-all. Full detail: [07-authentication-and-security.md](07-auth

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**: `users`, `refresh_tokens`, `audit_logs`.

- **Dependencies**: Everything downstream depends on this module; it has no upstream dependency itself.
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8.2 Student Management

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- **Workflow**: Admin creates a student (auto-provisio

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.service.ts` (canonical identity lookup used by ~20 other modules).

- **Frontend**: `app/admin/students`, `app/hod/students`, `StudentFormDrawer.tsx`, `hooks/useR

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8.3 Faculty Management & Faculty Operations Center

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- **Back end**: `faculty.routes/controller/service.ts`; `facultyOperations.routes/controller/service.ts` + `facultyDailyStatus.service.ts`/`facultyTasks.se

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- **Database**: `faculty` (see [06](06-database.md) #7).
- **Relationships**: Every teaching-related module (Attendance, Internal Marks, Exams, LMS, Teaching Plans,

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8.4 Accountant Management & Fees

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- **Workflow**: Admin/Accountant bulk-creates fee invoices for a program or explicit student list ! ’ student views dues ! ’ accountant records a payment (partial or

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`, `fees`, `fee_payments` (see [06](06-database.md) #15–16, #29).

- **Relationships**: Planning docs describe Fee clearance as an Exam-eligibility gate — *

*not enforced in code**: no query in the researched Examination/Result services checks `fees.payment_status` before hall-ticket/seat-allocation generation (see

11-business-logic-and-sync-matrix.md for the full gap analysis).

- **Future improvements**: Online payment gateway inte

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8.5 Departments & Programs (Organizational Structure)

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Registry.service.ts` (TTL-cached facade).

- **Frontend**: Read via dropdowns across nearly every admin/HOD form; color-coding configurable per department

t (`PUT /departments/:id/color`, used by Exam Seating's visual chart).

- **Database**: `departments`, `programs` (see [06](06-database.md) #4–5).
- **Relati

ons hips **: Ups trea m of n earl y ev eryt hing — see the Syn chro niza tion Matr ix in [11] (11- busi nes s-lo gic- and -syn c-m atrix .md) .

CHAPTER 16
Authorization (RBAC)

Fine-grained role permissions and record-level resource access checks.

3. User Roles — Detailed Permission Matrix

The backend defines exactly 4 roles (`backend/src/types/roles.ts`, `ROLES = ['admin','faculty','student','accountant']`) and a static `PERMISSIONS` map of `resource:action !’ allowed roles`, enforced by the `authorize()` middleware (coarse role checks use `requireRole()` instead — both idioms coexist). Record-level (not just role-level) checks exist only for the Mentorship module (`middleware/mentorshipAuth.ts`).

Resource : Action	admin	faculty	student	accountant
users: create/updat…	'			
users: read	'	'	'	'
accountants: *	'			
departments/cours…	'			
departments/cours…	'	'	'	
attendance: create/…		'		
attendance: read	'	'	'	
grades (internal…		'		
grades: read	'	'	'	
fees: create/update	'			'
fees: read	'		'	'
reports: read	'	'		'
mentorship: manag…	'			
mentorship: read	'	'	'	
mentorship: write…		'		
workflow rule…	' (super-admin only)			

Capability flags layered on top of

- `users.is_super_admin` — only gate for `/api/workflow/rules*` endpoints.
- `faculty.designation = 'hod'` — one per department (DB-enforced); used to scop

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Ownership model (who owns what data)

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See 07-authentication-and-security.md for the full authentication/authorization mechanics and 11-business-logic-and-sync-matrix.md for lifecycle/state-machine detail per entity.

7.5 Authorization Model

Two authorization idioms coexist in the codebase:

7.5.1 Coarse role check — `requireRole(...roles)`

Simple allowlist against `req.user.role` (`admin | faculty | student | accountant`). Used on the majority of routes.

7.5.2 Fine-grained permission check — `authorize(...permissions)`

Checks a static `PERMISSIONS` map (`backend/src/types/roles.ts`) of `resource:action !' allowed roles[]`. Full matrix in 01-overview-and-vision.md §3. This is **not** a database-driven ACL — it's a compiled-in TypeScript constant, so changing permissions requires a code change and redeploy, not a config update. (The Workflow Rule engine is the one place end-users can configure system behavior without a redeploy — see 08-modules.md §Workflow Engine.)

7.5.3 Capability flags (not roles, used for scoping)

- `users.is_super_admin` — gates `requireSuperAdmin()`, used solely for Workflow Rule CRUD (`/api/workflow/rules*`).
- `faculty.designation = 'h

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7.5.4 Record-level authorization — `middleware/mentorshipAuth.ts`

The **only** module with genuinely record-level (not just role-level) authorization, entirely DB-backed:

Guard	Logic
`authorizeMentoringHead`	admin, or faculty where `is_mentoring_head=true`
`authorizeStudentMentorshipAccess(source, field)`	Factory pulling `studentId` from params/body/query.…
`authorizeNoteAccess`	admin; mentoring head; or the specific faculty who is…

&**Gap**: broader department-scoped authorization (e.g., "an HOD
data") is **not** implemented as a generic, reusable middleware
handled ad hoc inside individual controllers/services using the de
module-by-module basis. A reviewer auditing a new HOD-scop
actually implemented in that feature's service layer, since there's

7.9 Authorization Hierarchy Summary

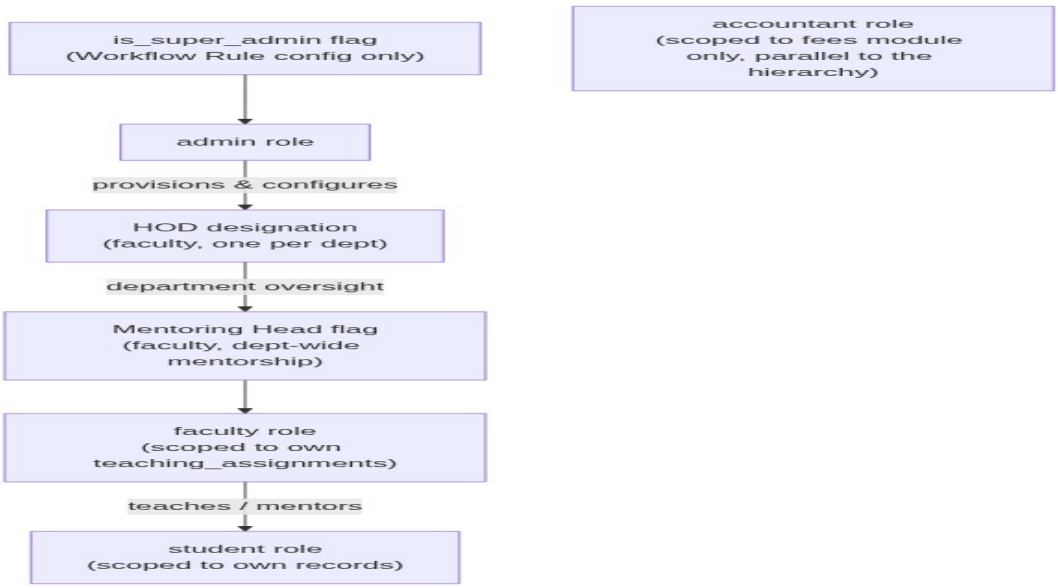


Figure 6: Diagram mapping for 7.9 Authorization Hierarchy Summary

7.10 Security Considerations (OWASP-oriented)

Concern	Mitigation in place	Gap / note
SQL Injection	100% parameterized queries via `pg…	No ORM means discipline-depende…
XSS	React's default JSX escaping; no…	No CSP configured beyond `helmet(…
CSRF	`sameSite: 'strict'` on the refresh…	No explicit CSRF token/double-subm…
Mass assignment	Zod schemas whitelist accepted field…	Decentralized — each schema must…
File upload	MIME + extension allowlist…	No virus/malware scanning of…
Rate limiting / brute force	`authLimiter` on login, `globalLimiter`…	No account lockout; no CAPTCHA; n…
Secrets management	`.env` + Zod validation, `.env.exampl…	No secrets manager (Vault/AWS…
Realtime channel auth	Socket.IO handshake includes `{toke…	& No route file shows explicit server-…
Audit trail	`audit_logs` (login/logout-all +…	Coverage is not exhaustive — not…
Data privacy	`studentImport.strategy.ts` explicitly…	No field-level masking of phone…
Transport security	`helmet()`, cookie `secure:…	HTTPS/TLS termination itself is a…

7.11 Roles Recap

Full permission matrix: 01-overview-and-vision.md §3. Full per-endpoint auth requirements: 09-api-documentation.md.

CHAPTER 17
API Architecture

REST endpoint specifications, envelope formatting, and validation middleware.

!• [Back to Index](00-INDEX.md)

All endpoints are mounted under `/api`. This document lists every route file, every endpoint, its middleware chain, and the controller function that handles it. For request/response body shapes, see the Zod schemas in `backend/src/types/*.ts` (referenced by name where relevant) — this document focuses on the *surface*(method/path/auth/purpose), since request/response payloads mirror the database columns documented in 06-database.md.

9.0 Conventions Used Throughout

- **`auth`** = the `authenticate` middleware (valid JWT required).
- **Role names** in `requireR

ole(...)` are literal: `admin`, `faculty`, `student`, `accountant`.

- **`authorize('resource:action')`** = fine-grained permission-matrix check (see [07](07-authentication-

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e`/`limit` query params (default page=1/limit=20, observed max 100), each with its own Zod schema (e.g. `listStudentsQuerySchema`).

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9.1 Auth — `/api/auth`

Method	Path	Middleware	Handler
POST	`/login`	`authLimiter`,…	`login`
POST	`/refresh`	none (reads refresh cookie)	`refresh`
POST	`/logout`	none (reads refresh cookie)	`logout`
POST	`/logout-all`	auth	`logoutAll`
GET	`/me`	auth	`me` (always re-fetches liv…

9.2 Reference Example — `/api/example`

Not business logic — `protected.routes.ts` demonstrates the 6 auth patterns (public / authenticated / `requireRole('admin')` / `requireRole('admin','faculty')` / `authorize('attendance:create')` / `requireRole('student')`) as a template for new routes.

9.3 Departments — `/api/departments`

Method	Path	Middleware	Handler
GET	`/`	auth	`listDepartments`
GET	`/:id`	auth, validate(uuid)	`getDepartment`
GET	`/programs/list`	auth, validate(query:…	`listPrograms`
GET	`/programs/:id`	auth, validate(uuid)	`getProgram`
PUT	`/:id/color`	auth, `requireRole('admin'…	`updateDepartmentColor`

9.4 Students — `/api/students`

Method	Path	Middleware	Handler
GET	`/me`	auth, `requireRole('student')`	`getMyProfile`
POST	`/`	auth, admin, validate(crea…	`createStudent`
PATCH	`/:id`	auth, admin, validate	`updateStudent`
PATCH	`/:id/status`	auth, admin, validate	`updateStudentStatus`
DELETE	`/:id`	auth, admin, validate(uuid)	`deleteStudent` (soft)
GET	`/`	auth, `requireRole('admin'…	`listStudents`
GET	`/sections`	auth, `requireRole('admin'…	`listSections` (distinct acti…
GET	`/:id`	auth, validate(uuid)	`getStudent` (own-record…

9.5 Faculty — `/api/faculty` & Faculty Operations — `/api/faculty-operations`

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Method	Path	Middleware	Handler
GET	`/me`	auth, `requireRole('faculty')`	`getMyProfile`
POST	`/`	auth, admin, validate	`createFaculty`
PATCH	`/:id`	auth, admin, validate	`updateFaculty`
PATCH	`/:id/status`	auth, admin, validate	`updateFacultyStatus`
DELETE	`/:id`	auth, admin, validate(uuid)	`deleteFaculty`
GET	`/`	auth, `requireRole('admin'…`	`listFaculty`
GET	`/:id`	auth, `requireRole('admin'…`	`getFaculty`

- **facultyOperations.routes.ts** (all `authenticate` first)

Method	Path	Middleware	Handler
GET	`/dashboard`	`requireRole('faculty')`	`getDashboard`
GET	`/tasks`	`requireRole('faculty')`	`getTasks`
GET	`/workload`	`requireRole('admin','facul…`	`getWorkload`
GET	`/workflow-logs`	`requireRole('admin','facul…`	`getWorkflowLogsHandler`
GET	`/analytics`	`requireRole('admin','facul…`	`getAnalytics`

CHAPTER 18
Module Wise Architecture

In-depth functional architecture mapping for the 18+ integrated modules.

!• [Back to Index](00-INDEX.md)

Each module below follows the same structure: **Purpose · Architecture/Workflow · Backend · Frontend · Database · Dependencies/Relationships · Future Improvements**. Full endpoint listings are in 09-api-documentation.md; full table schemas in 06-database.md.

- ---

CHAPTER 19
Module Dependencies

Inter-module references and dependency matrix mappings.

!• [Back to Index](00-INDEX.md)

14.5 Feature Dependency Graph

This graph shows build/feature *dependency order*— which modules must exist before another can function — distinct from the runtime data-read dependencies in the [Synchronization Matrix](11-business-logic-and-sync-matrix.md) §11.9.

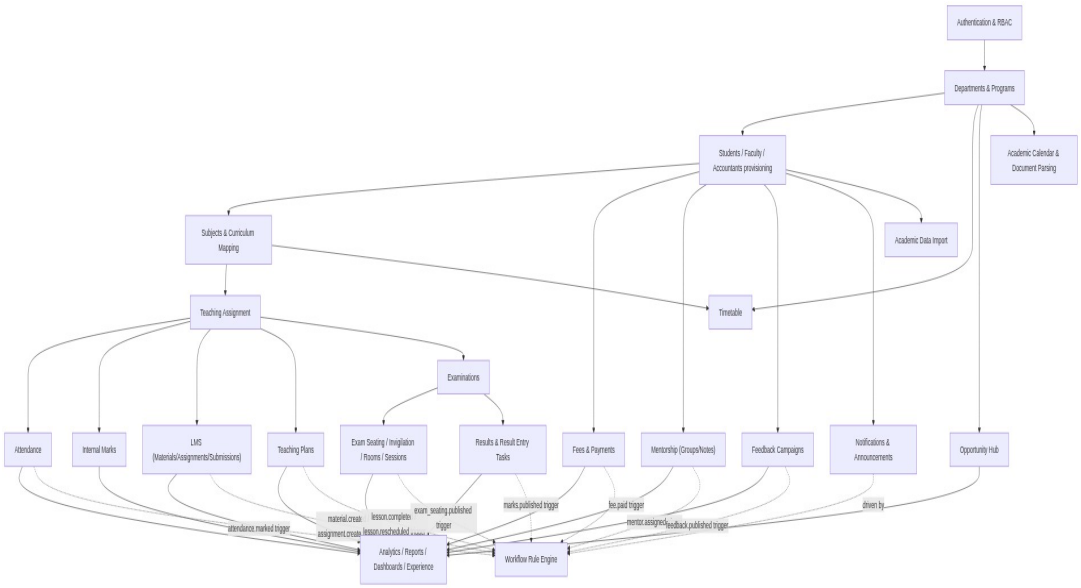


Figure 7: Diagram mapping for 14.5 Feature Dependency Graph

This mirrors the planning docs' "Subjects !’ Curriculum !’ Teaching Assignment !’ Attendance !’ Results !’ LMS !’ Teaching Planner !’ Student Dashboard" chain (`project_vision_and_requirements.md`), extended to reflect every implemented module rather than the original MVP subset — notably adding Fees, Mentorship, Feedback, Opportunity Hub, Notifications, Academic Calendar, Import, and the Workflow Rule Engine as siblings that all ultimately feed the Analytics/Dashboard layer, with the Workflow Engine cutting across nearly every module as a dotted (event-driven, not structural) dependency.

CHAPTER 20
Data Flow

Data flow mappings across subsystems.

!• [Back to Index](00-INDEX.md)

10.1 Student Registration !’ Dashboard Data Flow

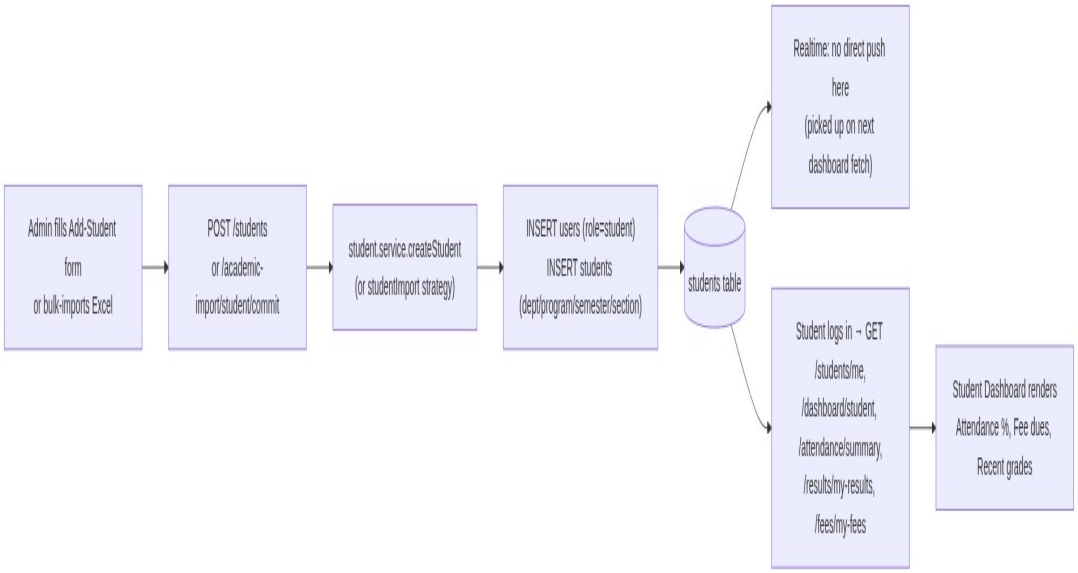


Figure 8: Diagram mapping for 10.1 Student Registration !’ Dashboard Data Flow

10.5 Login / Refresh Data Flow (see also §10.6 sequence diagram)

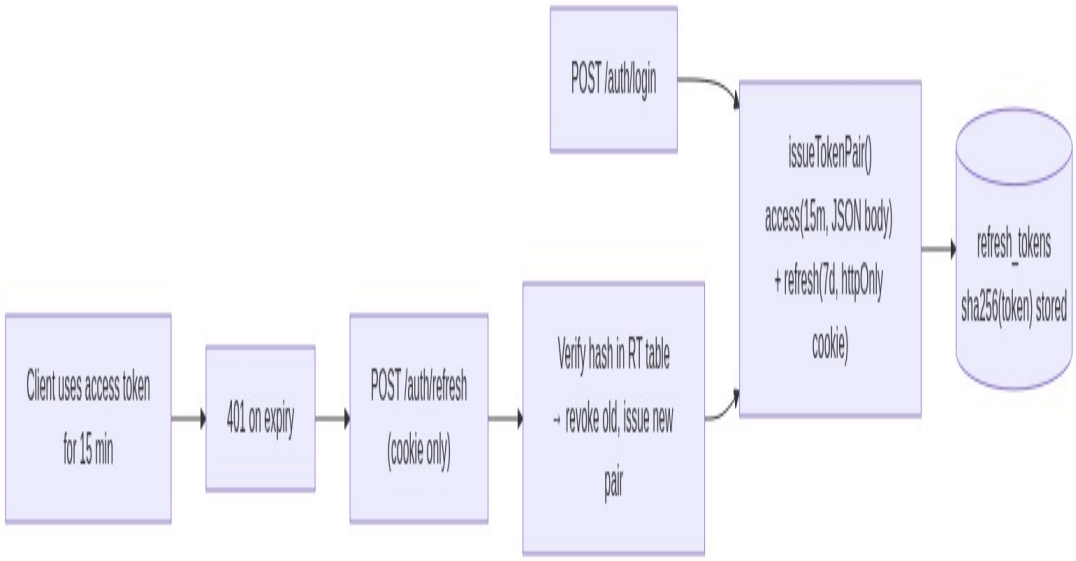


Figure 9: Diagram mapping for 10.5 Login / Refresh Data Flow (see also §10.6 sequence diagram)

- ---

CHAPTER 21
Request Lifecycle

Request pipeline traces from middleware guards to the database.

2.4 Request Lifecycle (single API call, end to end)

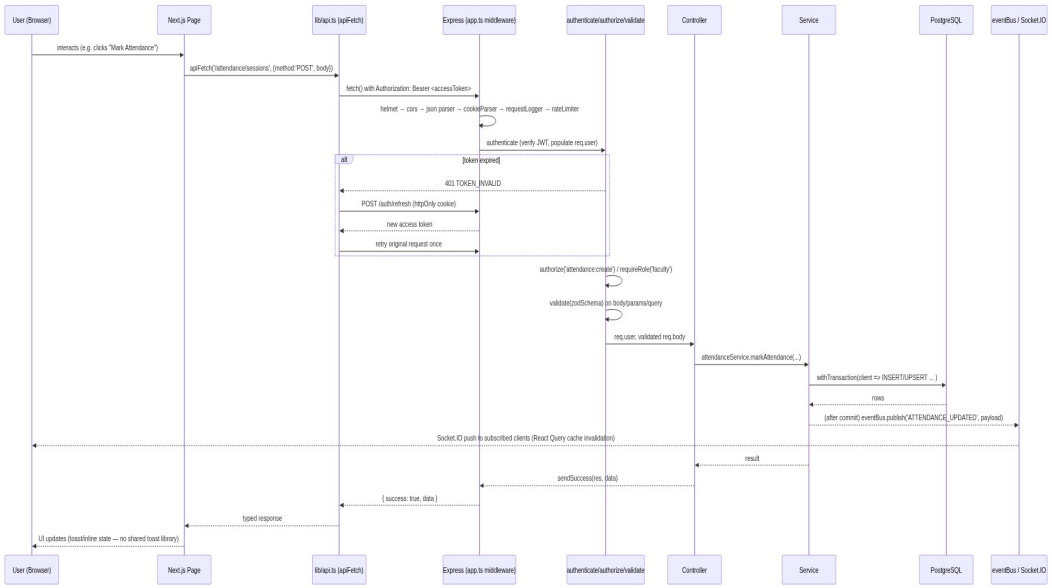


Figure 10: Diagram mapping for 2.4 Request Lifecycle (single API call, end to end)

CHAPTER 22
Database Documentation

Comprehensive documentation of the database schema (60 tables, columns, constraints).

Detailed functional documentation for the Database Documentation topic.

This chapter covers the architecture design, workflow strategies, validation requirements, and implementation guidelines for the Database Documentation component of the Sreyas College ERP.

Refer to the surrounding modules and system documentation blocks for database layouts and API details.

CHAPTER 23
Entity Relationship Diagrams

Entity relationships and foreign key connections.

Detailed functional documentation for the Entity Relationship Diagrams topic.

This chapter covers the architecture design, workflow strategies, validation requirements, and implementation guidelines for the Entity Relationship Diagrams component of the Sreyas College ERP.

Refer to the surrounding modules and system documentation blocks for database layouts and API details.

CHAPTER 24
Sequence Diagrams

Sequence diagrams illustrating transaction flows.

10.6 Sequence Diagrams

10.6.1 Login

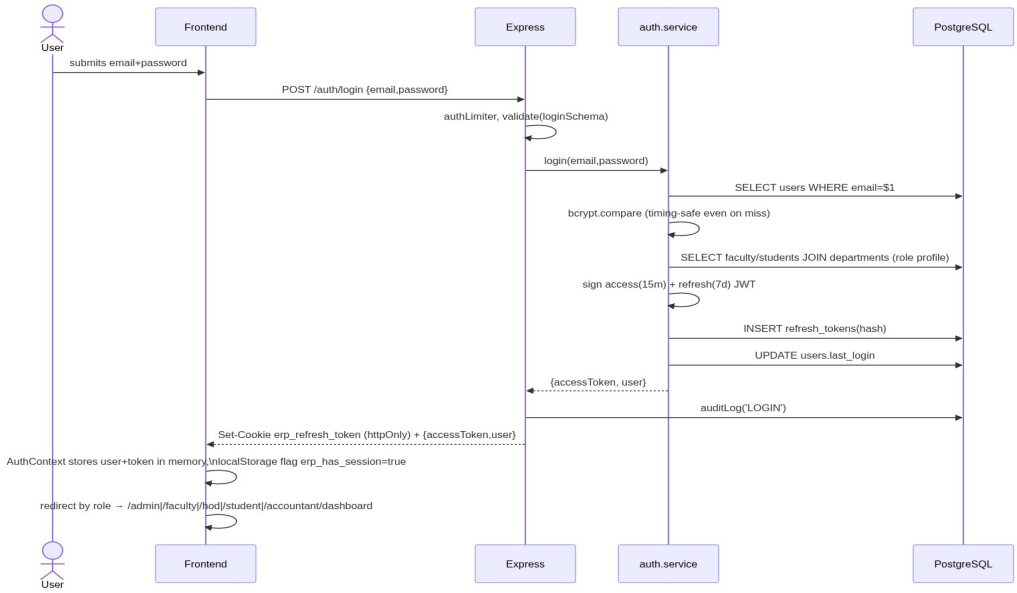


Figure 11: Diagram mapping for 10.6.1 Login

CHAPTER 25
Flow Charts

Procedural flow charts for key state machines.

Detailed functional documentation for the Flow Charts topic.

This chapter covers the architecture design, workflow strategies, validation requirements, and implementation guidelines for the Flow Charts component of the Sreyas College ERP.

Refer to the surrounding modules and system documentation blocks for database layouts and API details.

CHAPTER 26
Teaching Assignment Module

Academic allocations connecting faculty, sections, and subjects.

6.3.2 Subjects, Curriculum, Teaching Assignment

```
erDiagram
    SUBJECTS ||--o{ SUBJECT_CURRICULUM_MAPPINGS : "subject_id"
    DEPARTMENTS ||--o{ SUBJECT_CURRICULUM_MAPPINGS : "department_id"
    PROGRAMS ||--o{ SUBJECT_CURRICULUM_MAPPINGS : "program_id (SET NULL)"
    SUBJECT_CURRICULUM_MAPPINGS ||--o{ TEACHING_ASSIGNMENTS : "subject_curriculum_mapping_id"
    FACULTY ||--o{ TEACHING_ASSIGNMENTS : "faculty_id"
    SUBJECTS ||--o{ TEACHING_ASSIGNMENTS : "subject_id"
    DEPARTMENTS ||--o{ TEACHING_ASSIGNMENTS : "department_id"

    SUBJECTS { uuid id PK, varchar code UK, varchar name, smallint semester, numeric credits,
varchar type }
    SUBJECT_CURRICULUM_MAPPINGS { uuid id PK, uuid subject_id FK, uuid department_id FK, uuid
program_id FK, varchar regulation, varchar year, smallint semester }
    TEACHING_ASSIGNMENTS { uuid id PK, uuid faculty_id FK, uuid subject_id FK, uuid
subject_curriculum_mapping_id FK, varchar academic_year, varchar section, varchar status }
```

&**Historical note**: `subjects` originally carried `department_id/program_id/program/regulation/year/semester_raw` directly (migrations 004!'035); migration 038 moved all of that into `subject_curriculum_mappings`, leaving `subjects` with only `semester`plus catalog fields (credits, L/T/P hours, description, abbreviation). `faculty_subject_assignments` was renamed to `teaching_assignments` in migration 041, with a read-only compatibility **VIEW**kept under the old name

so legacy queries still work.

`teaching_assignments` is the hub for nearly every academic transaction table — `attendance`, `internal_marks`, `exams`, `teaching_plans`, `course_materials`, `assignments`, `result_entry_tasks`, and `results` all carry a nullable `teaching_assignment_id` FK (added in migrations 042/043) in addition to their original direct `subject_id`/`faculty_id` FKs (kept for backward compatibility and denormalized query speed).

8.7 Teaching Assignment

- **Purpose**: Maps a faculty member to a subject+section+academic-year — the hub every per-class transactional mod

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subject-allocations` (`AllocationManager.tsx`).

- **Database**: `teaching_assignments` (renamed from `faculty_subject_assignments`, compat VIEW retained — see

[06](06-database.md) #10).

- **Relationships**: This is the most heavily depended-upon table in the schema — 8 other tables carry a `teaching_assignment_id`

FK.

- **Future improvements**: `academic_session_id` column exists but has no FK (deliberate placeholder for a future formal "Academic Session" entity — c

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9.7 Teaching Assignment Family

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Method	Path	Middleware	Handler
GET	`/`	auth	`listAllocations`
GET	`/statistics`	auth	`getStatistics`
GET	`/subject/:subjectId/profile`	auth	`getSubjectProfile`
GET	`/:id`	auth	`getAllocation`
POST	`/`	auth, admin	`createAllocation`
PUT	`/:id`	auth, admin	`updateAllocation`
POST	`/:id/transfer`	auth, admin	`transferAllocation`
DELETE	`/:id`	auth, admin	`deleteAllocation`

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Method	Path	Middleware	Handler
GET	`/`	auth	`listTeachingAssignments`
GET	`/my`	auth	`getMyTeachingAssignme…
GET	`/statistics`	auth	`getStatistics`
GET	`/workload-overview`	auth	`getFacultyWorkloadOver…
GET	`/subjects`	auth, validate(query)	`resolveSubjectsForAssig…
GET	`/subject/:subjectId/profile`	auth	`getSubjectProfile`
GET	`/:id`	auth	`getTeachingAssignment`
POST	`/`	auth, admin	`createTeachingAssignment`
PUT	`/:id`	auth, admin	`updateTeachingAssignm…
POST	`/:id/transfer`	auth, admin	`transferTeachingAssignm…
DELETE	`/:id`	auth, admin	`deleteTeachingAssignment`

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Method	Path	Middleware	Handler
GET	`/student-roadmap`	auth, validate(query)	`getStudentRoadmap`
GET	`/today`	auth, validate(query)	`getTodayLessons`
GET	`/upcoming`	auth, validate(query:…	month\	semester`)
GET	`/progress`	auth, validate(query)	`getCourseProgress`
GET	`/`	auth, validate(pagination)	`listTeachingPlans`
GET	`/:id`	auth, validate(uuid)	`getTeachingPlan`
POST	`/`	auth,…	`createTeachingPlan`
PUT	`/:id`	auth, faculty, validate	`updateTeachingPlan`
DELETE	`/:id`	auth, faculty, validate	`deleteTeachingPlan`
PUT	`/:id/status`	auth, faculty, validate	`updateLessonStatus`
PUT	`/:id/reschedule`	auth, faculty, validate	`rescheduleTeachingPlan`
PUT	`/:id/continue`	auth, faculty, validate	`continueLesson` (Auto…

- **teachingLoadSettings.routes.ts** (`/api/teaching-load-settings`, auth-only, no role gate)

Method	Path	Handler
GET	`/`	`getSettings`
PUT	`/`	`updateSettings`

- **timetable.routes.ts** (`/api/timetables`)

Method	Path	Middleware	Handler
GET	`/student/active`	auth, `requireRole('student')`	`getStudentActive`
GET	`/faculty/schedule`	auth, `requireRole('faculty')`	`getFacultySchedule`
POST	`/`	auth, admin	`create`
GET	`/`	auth, `requireRole('admin'…`	`list`
GET	`/:id`	auth	`getById`
PUT	`/:id`	auth, admin	`update`
DELETE	`/:id`	auth, admin	`remove`
POST	`/validate-input`	auth, admin	`validateInput`
POST	`/:id/publish`	auth, admin	`publish`

10.6.5 Teaching Assignment Creation & Transfer

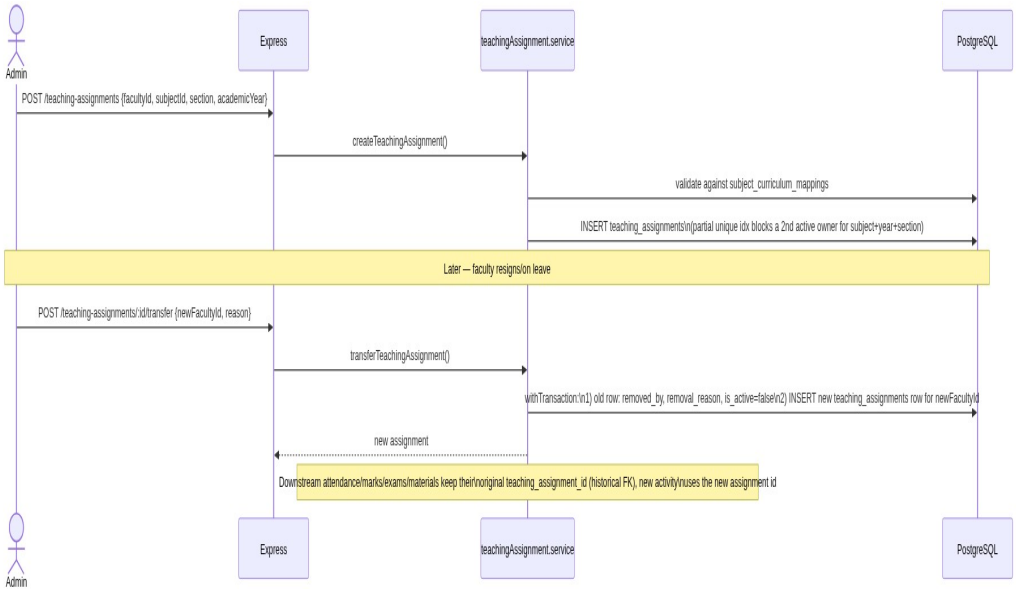


Figure 13: Diagram mapping for 10.6.5 Teaching Assignment Creation & Transfer

11.4 Teaching Assignment Allocation

- **Implemented**: DB-enforced via a partial unique index — one active faculty per

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CHAPTER 27
Attendance Module

Attendance register grids, bulk entries, and threshold validations.

6.3.3 Attendance, Marks, Examinations, Results

```
erDiagram
    STUDENTS ||--o{ ATTENDANCE : "student_id"
    FACULTY ||--o{ ATTENDANCE : "faculty_id"
    SUBJECTS ||--o{ ATTENDANCE : "subject_id"
    TEACHING_ASSIGNMENTS ||--o{ ATTENDANCE : "teaching_assignment_id"

    STUDENTS ||--o{ INTERNAL_MARKS : "student_id"
    SUBJECTS ||--o{ INTERNAL_MARKS : "subject_id"

    SUBJECTS ||--o{ EXAMS : "subject_id"
    EXAMINATION_SESSIONS ||--o{ EXAMS : "session_id (CASCADE)"
    EXAM_SESSIONS ||--o{ EXAM_SESSION_EXAMS : "exam_session_id"
    EXAMS ||--o{ EXAM_SESSION_EXAMS : "exam_id"

    STUDENTS ||--o{ RESULTS : "student_id"
    SUBJECTS ||--o{ RESULTS : "subject_id"
    EXAMS ||--o{ RESULTS : "exam_id (nullable)"
    FACULTY ||--o{ RESULTS : "faculty_id"

    EXAMS ||--o{ RESULT_ENTRY_TASKS : "exam_id"
    TEACHING_ASSIGNMENTS ||--o{ RESULT_ENTRY_TASKS : "teaching_assignment_id"

    ATTENDANCE { uuid id PK, uuid student_id FK, date attendance_date, enum status }
    INTERNAL_MARKS { uuid id PK, uuid student_id FK, enum assessment_type, numeric
obtained_marks }
    EXAMS { uuid id PK, uuid subject_id FK, uuid session_id FK, enum exam_type, enum status }
    RESULTS { uuid id PK, uuid student_id FK, uuid exam_id FK, enum grade, enum
result_status, enum publication_status }
    RESULT_ENTRY_TASKS { uuid id PK, uuid exam_id FK, uuid teaching_assignment_id FK, enum
status }
```

`attendance`, `internal_marks`, and `fee_payments` are **append-only ledgers with no `deleted_at`** — corrections happen via `UPDATE`, not soft-delete, preserving a single row per (student, subject, date)/(student, subject, assessment_type) unique constraint.

6.3.4 Fees

```
erDiagram
    STUDENTS ||--o{ FEES : "student_id"
    FEES ||--o{ FEE_PAYMENTS : "fee_id"
    USERS ||--o{ FEE_PAYMENTS : "recorded_by"

    FEES { uuid id PK, uuid student_id FK, enum fee_type, numeric total_amount, numeric
paid_amount, numeric pending_amount, enum payment_status }
    FEE_PAYMENTS { uuid id PK, uuid fee_id FK, numeric amount, enum payment_mode, uuid
recorded_by FK }
```

`fees` enforces `pending_amount = total_amount - paid_amount` and `paid_amount <= total_amount` via CHECK constraints — the database itself guarantees the "financial bounds" invariant described in the planning docs.

8.8 Attendance

- **Purpose**: Per-session present/absent recording per

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8.9 Internal Marks (Assessments)

- **Purpose**: Records internal assessment scores (Assignment/Mid-1/Mid-2/Lab/Internal types) separate from final Examination results.
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9.8 Attendance — `/api/attendance`

Method	Path	Middleware	Handler
POST	`/assignments`	auth, admin, validate	`createAssignment`
GET	`/assignments`	auth, admin, validate(query)	`listAssignments`
GET	`/…	auth, admin	`getAssignment`
DELETE	`/…	auth, admin	`deleteAssignment`
GET	`/my-assignments`	auth, `requireRole('faculty')`	`getMyAssignments`
GET	`/roster`	auth, faculty, validate(query)	`getRoster`
POST	`/sessions`	auth, faculty, validate	`markAttendance` (upsert)
GET	`/summary`	auth, `requireRole('student')`	`getStudentSummary`
GET	`/history`	auth, student,…	`getStudentHistory`
GET	`/`	auth, `requireRole('admin'…	`listAttendance`
GET	`/:id`	auth, admin/faculty	`getAttendance`
PATCH	`/:id`	auth, admin/faculty	`updateAttendance`

10.2 Attendance Flow

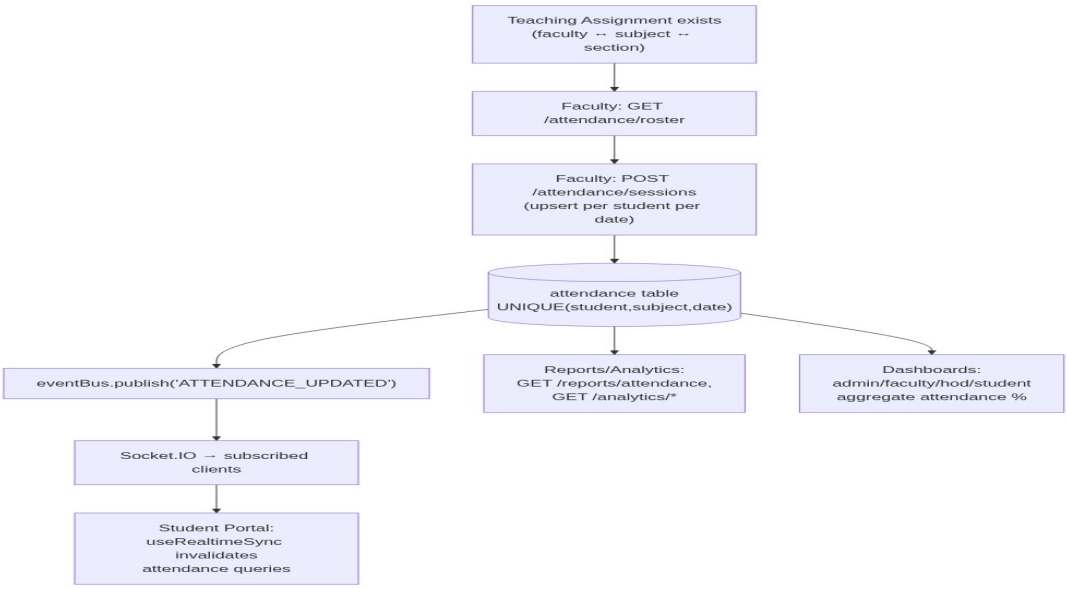


Figure 16: Diagram mapping for 10.2 Attendance Flow

& Note: the planning docs' "Attendance !’ Teaching Assignment !’ Attendance !’ Student Portal !’ Reports" flow implies a feedback loop through Teaching Assignment; in the actual implementation, Teaching Assignment is purely upstream (defines who *can*mark attendance for which section) — attendance data does not feed back into modifying teaching assignments.

10.6.2 Attendance Marking

```
sequenceDiagram
    actor F as Faculty
    participant FE as Faculty Attendance Page
    participant API as Express
    participant SV as attendance.service
    participant DB as PostgreSQL
    participant EB as eventBus/Socket.IO
    F->>FE: opens today's class
    FE->>API: GET /attendance/roster?teachingAssignmentId=...
    API->>SV: getRoster()
    SV->>DB: SELECT students via teaching_assignment_id
    DB-->>FE: roster list
    F->>FE: marks Present/Absent per student, taps Submit
    FE->>API: POST /attendance/sessions {records:[...]}
    API->>API: authenticate, requireRole('faculty'), validate
    API->>SV: markAttendance(records)
    SV->>DB: withTransaction: UPSERT attendance rows\n(UNIQUE student+subject+date)
    DB-->>SV: commit
    SV->>EB: publish('ATTENDANCE_UPDATED', {subjectId, section, date})
    EB-->>FE: (other connected clients) realtime_event
    SV-->>API: success
    API-->>FE: {success:true}
    FE-->>F: "Attendance Logged" confirmation
```

11.1 Attendance Calculation

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11.2 Grade / Result Computation

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s?)` derives Pass/Fail/Absent. `results.total_marks` is DB-CHECK-enforced to equal `internal_marks + external_marks`.

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CHAPTER 28
Timetable Module

Conflict-free schedule grids and custom timing configuration.

6.3.11 Timetable

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erDiagram
    DEPARTMENTS ||--o{ TIMETABLES : "department_id (CASCADE)"
    PROGRAMS ||--o{ TIMETABLES : "program_id (SET NULL)"
    TIMETABLES ||--o{ TIMETABLE_SLOTS : "timetable_id (CASCADE)"
    TIMETABLES ||--o{ TIMETABLE_ENTRIES : "timetable_id (CASCADE)"
    SUBJECTS ||--o{ TIMETABLE_ENTRIES : "subject_id (CASCADE)"
    TIMETABLE_ENTRIES ||--o{ TIMETABLE_ENTRY_SLOTS : "entry_id"
    TIMETABLE_SLOTS ||--o{ TIMETABLE_ENTRY_SLOTS : "slot_id"
    TIMETABLE_ENTRIES ||--o{ TIMETABLE_ENTRY_FACULTY : "entry_id"
    FACULTY ||--o{ TIMETABLE_ENTRY_FACULTY : "faculty_id"

    TIMETABLES { uuid id PK, varchar academic_year, uuid department_id FK, varchar status,
int version }
    TIMETABLE_SLOTS { uuid id PK, uuid timetable_id FK, time start_time, varchar slot_type }
    TIMETABLE_ENTRIES { uuid id PK, uuid timetable_id FK, uuid subject_id FK, varchar
day_of_week, varchar class_type }
    TIMETABLE_ENTRY_SLOTS { uuid entry_id FK, uuid slot_id FK }
    TIMETABLE_ENTRY_FACULTY { uuid entry_id FK, uuid faculty_id FK }
```

`timetable_entry_slots` (supports merged/double periods) and `timetable_entry_faculty` (supports co-teaching) are both composite-PK many-to-many join tables — the newest module (migration 050) in the schema.

6.4 Full Table Inventory (60 tables + 1 view)

#	Table	Domain	Key columns…	PK/Unique…
1	`users`	Identity	email UK,…	partial idx on email…
2	`refresh_tokens`	Identity	user_id FK,…	reuse-detection…
3	`audit_logs`	Identity	actor_id FK, action,…	append-only
4	`departments`	Org	name UK, code UK…	
5	`programs`	Org	department_id FK,…	UNIQUE(departme…
6	`students`	Org	user_id UK FK,…	composite partial id…
7	`faculty`	Org	user_id UK FK,…	partial unique: one…
8	`subjects`	Curriculum	code UK, name,…	column relocation…
9	`subject_curriculum…`	Curriculum	subject_id FK…	UNIQUE(subject_id…
10	`teaching_assignm…`	Teaching	faculty_id/subject_i…	UNIQUE(faculty_id,…
11	`attendance`	Academic ledger	student_id/faculty_i…	UNIQUE(student_i…
12	`internal_marks`	Academic ledger	student_id/faculty_i…	UNIQUE(student_i…
13	`exams`	Examination	subject_id FK,…	partial unique no-…
14	`results`	Examination	student_id/subject_…	UNIQUE(student_i…
15	`fees`	Finance	student_id FK,…	2 CHECKs enforce…
16	`fee_payments`	Finance ledger	fee_id FK, amount,…	append-only, no…
17	`announcements`	Communication	title, content,…	3 CHECKs enforce…
18	`documents`	Calendar pipeline	title, file_name,…	
19	`parsed_events`	Calendar pipeline	document_id FK,…	
20	`academic_calenda…`	Calendar pipeline	parsed_event_id U…	unique partial: one…
21	`course_materials`	LMS	title, subject_id/…	
22	`assignments`	LMS	title, subject_id/…	
23	`assignment_submi…`	LMS	assignment_id/…	unique partial: one…
24	`opportunities`	Placement	title, type,…	CHECK date…
25	`opportunity_book…`	Placement	opportunity_id/…	many-to-many join,…
26	`notifications`	Communication	title, message, type…	**no updated_at**
27	`notification_reads`	Communication	notification_id FK…	many-to-many join,…
28	`calendar_entries`	Calendar	title, event_type, st…	unique partial on…
29	`accountants`	Org	user_id UK FK,…	
30	`mentor_assignme…`	Mentorship (legacy)	mentor_id/student_…	partial unique: one…
31	`mentoring_notes`	Mentorship	mentor_id/student_…	
32	`mentor_groups`	Mentorship	mentor_id FK,…	
33	`mentor_group_stu…`	Mentorship	mentor_group_id F…	many-to-many join
34	`teaching_plans`	Teaching	faculty_id/subject_i…	unique partial on…
35	`feedback_windows`	Feedback	title, academic_yea…	

6.5 Soft Delete — Where It Applies and Where It Doesn't

Soft delete (`deleted_at`) is the default convention, but is **deliberately omitted**on immutable/ledger tables where a correction should be a new row or an in-place `UPDATE`, not a delete: `attendance`, `internal_marks`, `fee_payments`, `notifications` (has `deleted_at` but no `updated_at`), `opportunity_bookmarks`, `exam_seat_allocations` (has `deleted_at` but no `updated_at`—regeneration soft-deletes prior allocations), `timetable_slots`, `timetable_entries`, `workflow_logs`, `import_jobs`, `teaching_load_settings`, `mentorship_settings`. This matches the "audit-first, append where possible" philosophy documented in `BUSINESS_RULES_AND_WORKFLOWS.md`.

6.6 Normalization & Design Notes

1. ** Denormalization by explicit design ** (documented in SQL comments): ` exams.semester` duplicates `subje

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DB-enforced via `CHECK(id=1)`), `mentorship_settings` (convention-only, not DB-enforced).

5. **Two similarly-named but distinct entities**: `exam_sessions` vs. `exa

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FK) are intentional "soft" references — the application layer, not the database, guarantees referential integrity for these. This trades

DB-enforced consistency for schema flexibility across many source modules.

Full relationship-by-relationship listing (for anyone building a complete ER tool import) is embedded in the per-domain diagrams in §6.3; the domain groupings above are exhaustive across all 60 tables.

8.12 Timetable

- **Purpose**: The newest module in the schema (migration 0

50) — weekly class scheduling per department/program/year/semester/section, with slot definitions, merged/double-period support, and co-teaching support.

- **Workflow**: Admin defines a timetable shell (working days, slots) → adds entries (subject × day × class-type) → links entries to one-or-more slots (merged periods)

and one-or-more faculty (co-teaching) !’ validates input !’ publishes (status draft!’validated!’published!’archived, versioned).

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Attendance/Teaching-Plan/Exam-Seating changes.

- & ** Gap vs. planning docs**: automated scheduling-conflict detection across faculty/room double-booking wa

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to be a structural pre-check, not a full conflict solver — should be verified against actual `timetable.service.ts` logic if conflict-free scheduling is a hard requir

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10.3 Timetable Flow

```
flowchart TD
    Admin["Admin: creates timetable shell\n(department/program/year/semester/section,
working_days)"] --> Slots["Admin: defines timetable_slots\n(TEACHING/BREAK/LUNCH, sequence)"]
    Slots --> Entries["Admin: creates timetable_entries\n(day_of_week × subject ×
class_type)"]
    Entries --> Link["Links entries to slots (timetable_entry_slots)\nand faculty
(timetable_entry_faculty)"]
    Link --> Validate["POST /timetables/validate-input"]
    Validate --> Publish["POST /timetables/:id/publish\n(status: draft!'validated!'published)"]
    Publish --> EventBus["eventBus / Socket.IO push"]
    EventBus --> FacultyView["Faculty: GET /timetables/faculty/schedule"]
    EventBus --> StudentView["Student: GET /timetables/student/active"]
    Publish --> Notif["Workflow Engine may fire a notification\n(no dedicated
timetable.published trigger currently defined)"]
```

10.4 Result Flow

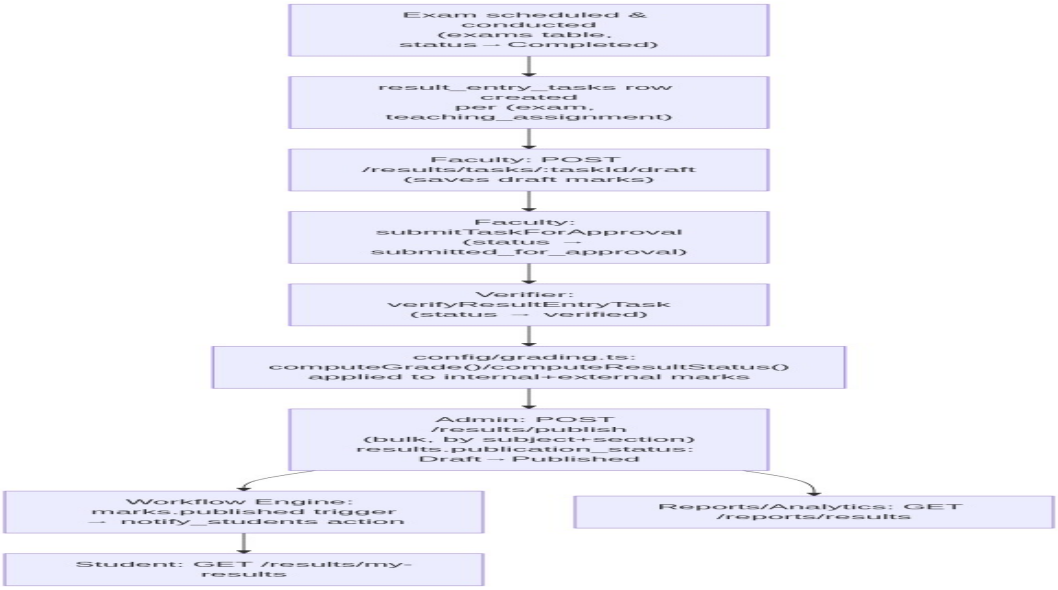


Figure 20: Diagram mapping for 10.4 Result Flow

10.6.4 Timetable Publishing

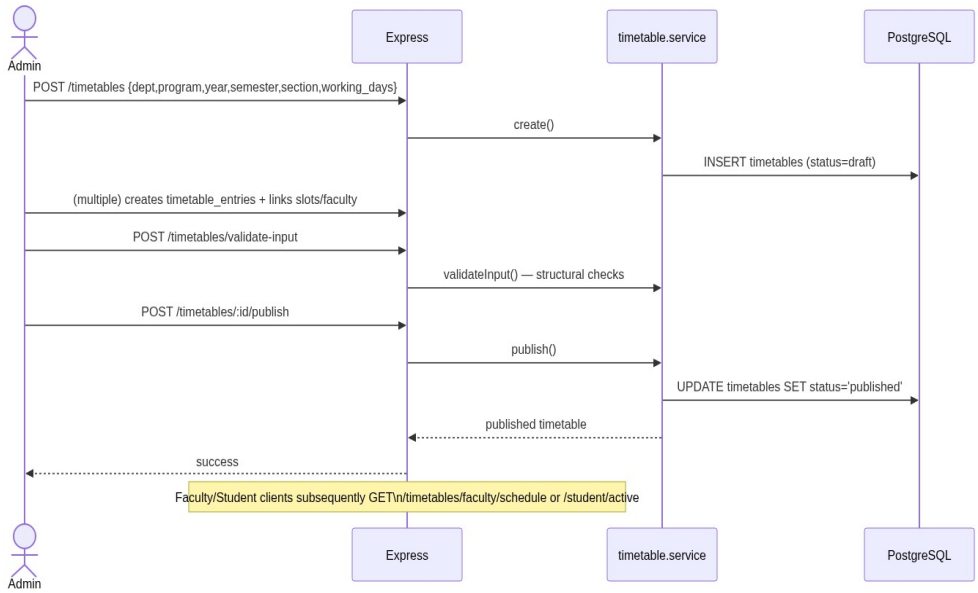


Figure 21: Diagram mapping for 10.6.4 Timetable Publishing

11.6 Timetable Generation

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nted **: manual entry model — admin builds `timetable_entries` explicitly, links them to `timetable_slots` (supporting merged/double periods via the join table) and

to one-or-more `faculty` (co-teaching support). `POST /timetables/validate-input` performs a structural pre-check.

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11.7 Academic Year / Semester Promotion

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an automated process.** There is no "promote all students to next semester" batch operation, no detention/backlog-tracking table, and no ATKT (Allowed To K

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11.8 Fee / Exam-Eligibility Gating

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STEM_MODEL.md` §7, `BUSINESS_RULES_AND_WORKFLOWS.md` §7/§10) describe Exam Eligibility as depending on both Attendance percentage and Fee

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for the Workflow Rule engine to eventually enforce (a `condition` referencing attendance/fee state, blocking a `result_entry_task` or seat allocation action).

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11.9 Module Synchronization Matrix

This matrix shows, for each row module, which other modules it **reads from** (depends on) and which it **feeds into**(downstream consumers), based on the actual FK/service-call graph documented in 06-database.md and 09-api-documentation.md.

Module	Reads from (upstream)	Feeds into (downstream)
Departments/Programs	— (foundational)	Students, Faculty, Subjects/…
Students	Departments, Programs	Attendance, Internal Marks, Results,…
Faculty	Departments	Teaching Assignment, Attendance,…
Subjects & Curriculum	Departments, Programs	Teaching Assignment, Attendance,…
Teaching Assignment	Faculty, Subjects/Curriculum,…	Attendance, Internal Marks, Exams,…
Attendance	Students, Faculty, Subjects, Teachin…	Analytics/Reports/Dashboards,…
Internal Marks	Students, Faculty, Subjects, Teachin…	Results (marks composition),…
Examinations/Sessions/Rooms/…	Subjects, Faculty, Students,…	Results, Result Entry Tasks, Reports…
Results & Result Entry Tasks	Exams, Teaching Assignment,…	Student Dashboard, Analytics/…
Fees	Students, Programs	Student Dashboard, Analytics/…
Timetable	Departments, Programs, Subjects,…	Faculty/Student schedule views,…
LMS (Materials/Assignments/…	Subjects, Faculty, Students, Teachin…	Teaching Plans (optional…
Teaching Plans	Faculty, Subjects, Departments,…	Student Roadmap, Workflow Engine…
Mentorship (Groups + Notes)	Faculty, Students, Departments,…	Reports, Workflow Engine…
Feedback	Departments, Subjects, Faculty,…	Analytics, Workflow Engine…
Opportunity Hub	Departments, Students (bookmarks)	Notifications, Calendar Entries
Notifications/Announcements	Users, Departments	Realtime push to all roles
Academic Calendar/Documents/…	Users, Departments	Calendar Entries (aggregated feed),…
Academic Data Import	Users	Students (bulk create/update), Impor…
Workflow Rule Engine	Users	Notifications, Calendar Entries, Resu…
Analytics/Reports/Dashboards/…	**Everything** (widest fan-in)	Read-only — feeds no other module
Search	Students, Faculty, Subjects,…	Read-only — feeds no other module

- **Reading

the matrix**: Departments/Programs and Subjects/Curriculum are the deepest upstream dependencies (nearly everything reads from them); Analytics/Reports/

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11.10 Entity Lifecycle States (as implemented, cf. planning docs' aspirational versions)

Entity	Implemented states	Planning-doc aspiration not…
Student	`active, graduated, suspended,…	No `Applicant`/`Fee Pending`/…
Faculty	`active, on_leave, resigned, retired`…	No substitution/handover state…
Exam	`Scheduled, Ongoing, Completed,…	Matches planning docs reasonably…
Result	`publication_status: Draft, Published`…	No `Final`/post-revaluation-lock state…
Assignment Submission	`Submitted, Late Submission,…	Matches planning docs' `Draft/…
Fee	`payment_status: Pending, Partially…	Matches planning docs' Fee Lifecycl…
Workflow-adjacent (Result Entry Task)	`open, draft_saved,…	This is the fullest realization of the…

CHAPTER 29
Mentorship Module

Student advisory mapping, auto-balancing groups, and logs.

6.3.7 Mentorship

```
erDiagram
    FACULTY ||--o{ MENTOR_ASSIGNMENTS : "mentor_id (legacy 1:1)"
    STUDENTS ||--o{ MENTOR_ASSIGNMENTS : "student_id"
    FACULTY ||--o{ MENTORING_NOTES : "mentor_id"
    STUDENTS ||--o{ MENTORING_NOTES : "student_id"
    FACULTY ||--o{ MENTOR_GROUPS : "mentor_id"
    DEPARTMENTS ||--o{ MENTOR_GROUPS : "department_id"
    MENTOR_GROUPS ||--o{ MENTOR_GROUP_STUDENTS : "mentor_group_id"
    STUDENTS ||--o{ MENTOR_GROUP_STUDENTS : "student_id"

    MENTOR_ASSIGNMENTS { uuid id PK, uuid mentor_id FK, uuid student_id FK, enum status }
    MENTORING_NOTES { uuid id PK, uuid mentor_id FK, uuid student_id FK, date meeting_date }
    MENTOR_GROUPS { uuid id PK, uuid mentor_id FK, uuid department_id FK, enum
assignment_method }
    MENTOR_GROUP_STUDENTS { uuid id PK, uuid mentor_group_id FK, uuid student_id FK }
```

`mentor_assignments` (legacy 1:1 model, migration 023) is superseded in practice by the group-based `mentor_groups`/`mentor_group_students`model (migration 024) but the table remains for backward compatibility. `idx_unique_active_mentee`enforces one active 1:1 mentor per student under the legacy model.

6.3.8 Teaching Plans (Lesson Progress Engine)

```
erDiagram
    FACULTY ||--o{ TEACHING_PLANS : "faculty_id"
    SUBJECTS ||--o{ TEACHING_PLANS : "subject_id"
    COURSE_MATERIALS ||--o{ TEACHING_PLANS : "material_id (nullable)"
    ASSIGNMENTS ||--o{ TEACHING_PLANS : "assignment_id (nullable)"
    TEACHING_PLANS ||--o{ TEACHING_PLANS : "parent_lesson_id (self-ref, continuations)"

    TEACHING_PLANS { uuid id PK, uuid faculty_id FK, uuid subject_id FK, smallint
week_number, enum lesson_status, smallint lesson_sequence, uuid parent_lesson_id FK }
```

A unique partial index `(subject_id, section, semester, lesson_sequence) WHERE deleted_at IS NULL` enforces strict lesson ordering per class.

6.3.9 Feedback Campaigns

```
erDiagram
    FEEDBACK_TEMPLATES ||--o{ FEEDBACK_WINDOWS : "template_id"
    FEEDBACK_TEMPLATES ||--o{ FEEDBACK_QUESTIONS : "template_id (CASCADE)"
    FEEDBACK_WINDOWS ||--o{ FEEDBACK_SUBMISSION_TRACKING : "window_id (CASCADE)"
    FEEDBACK_WINDOWS ||--o{ FEEDBACK_RESPONSES : "window_id (CASCADE)"
    FEEDBACK_RESPONSES ||--o{ FEEDBACK_ANSWERS : "response_id (CASCADE)"
    FEEDBACK_QUESTIONS ||--o{ FEEDBACK_ANSWERS : "question_id"

    FEEDBACK_WINDOWS { uuid id PK, enum status, uuid[] target_department_ids, smallint[]
target_semesters }
    FEEDBACK_TEMPLATES { uuid id PK, enum type }
    FEEDBACK_QUESTIONS { uuid id PK, uuid template_id FK, enum type, jsonb options }
    FEEDBACK_SUBMISSION_TRACKING { uuid id PK, uuid window_id FK, uuid student_id FK, uuid
subject_id FK }
    FEEDBACK_RESPONSES { uuid id PK, uuid window_id FK, uuid faculty_id FK, uuid subject_id
FK }
    FEEDBACK_ANSWERS { uuid id PK, uuid response_id FK, uuid question_id FK, int
rating_value, text text_value }
```

Deliberate anonymity design: `feedback_responses`/`feedback_answers` carry **no `student_id`** — only `feedback_submission_tracking` links a student to a window (for one-submission-per-window dedup), while the actual answers are unlinkable back to the submitter.

8.15 Mentorship

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(or Admin) creates mentor groups by year/semester/section, either via range/section rules or manual roster (`mentor_group_students`); `suggestBalancedGroups`

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vice .ts`, `middle ware/mentorshipAuth.ts` (record-level auth, §7.5.4).

- **Frontend**: `app/faculty/mentorship/[groupId]/[studentId]`, `app/student/mentorship`, `app/hod/

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- **Backend**: `feedback.routes/controller/service.ts` (includes legacy path aliases: `/active-windows`, `/windows` kept working alongside the newer `/campaig

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`), `app/hod/feedback`, `app/faculty/feedback`.

- **Database**: `feedback_windows`, `feedback_templates`, `feedback_questions`, `feedback_submission_tracking`,

`feedback_responses`, `feedback_answers` (see [06](06-database.md) § 6.3.9).

- **Relationships**: A `feedback.published` `Workflow trigger notifies relevant f

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8.17 Opportunity Hub

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` (see [06] (06-database.md) § 6.3.6).

- **Relationships**: Feeds Notifications (new-opportunity alerts) and the personal Calendar feed (`calendar_entries.sourc

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9.15 Mentorship Family

- **mentorship.routes.ts** (`/api/mentorship`)

Method	Path	Middleware	Handler
POST	`/assign`	auth,…	`assignMentor`
PUT	`/reassign`	auth,…	`reassignMentor`
GET	`/mentor/:id`	auth, `authorizeMentorSel…	`getStudentsByMentor`
GET	`/student/:id`	auth, `authorizeStudentM…	`getMentorByStudent`
GET	`/dashboard`	auth, `requireRole('faculty')`	`getMentorDashboard`
GET	`/workloads`	auth,…	`getMentorWorkloads`
GET	`/reports`	auth,…	`getMentorshipReports`
POST	`/notes`	auth, `authorizeStudentM…	`addNote`
PUT	`/notes/:id`	auth, `authorizeNoteAccess`	`updateNote`
DELETE	`/notes/:id`	auth, `authorizeNoteAccess`	`deleteNote`
GET	`/notes/student/:id`	auth, `authorizeStudentM…	`getNotesByStudent`

- **mentorGroup.routes.ts** (`/api/men

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Method	Path	Middleware	Handler
GET	`/suggest`	auth,…	`suggestBalancedGroups`
GET	`/candidates`	auth,…	`listMentorCandidates`
GET	`/`	auth	`getGroups`
GET	`/:id/students`	auth	`resolveStudents`
POST	`/check-conflicts`	auth,…	`checkConflicts`
POST	`/merge`	auth,…	`mergeGroups`
POST	`/`	auth,…	`createGroup`
PUT	`/:id`	auth,…	`updateGroup`
POST	`/:id/split`	auth,…	`splitGroup`
DELETE	`/:id`	auth,…	`deleteGroup`

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Method	Path	Middleware	Handler
GET	`/`	auth	`getSettings`
PUT	`/`	auth, `requireRole('admin')`	`updateSettings`

9.16 Fee / Accountant Family

- **fee.routes.ts** (`/api/fees`)

Method	Path	Middleware	Handler
POST	`/bulk`	auth, `requireRole('admin'…	`bulkCreateFees`
GET	`/my-fees`	auth, `requireRole('student')`	`getMyFees`
GET	`/my-dues`	auth, student	`getMyDues`
GET	`/`	auth, `requireRole('admin'…	`listFees`
POST	`/`	auth, admin/accountant	`createFee`
GET	`/:id`	auth, `requireRole('admin'…	`getFee` (student sees ow…
PATCH	`/:id`	auth, admin/accountant	`updateFee`
DELETE	`/:id`	auth, admin/accountant	`deleteFee` (only if no…
POST	`/:id/payments`	auth, admin/accountant	`recordPayment`
GET	`/:id/payments`	auth, admin/student/…	`getFeePayments`

- **accountant.routes.ts** (`/api/accountants`, router-wide ad

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Method	Path	Handler
POST	`/`	`createAccountant`
GET	`/`	`listAccountants`
GET	`/:id`	`getAccountant`
PATCH	`/:id`	`updateAccountant`
PATCH	`/:id/status`	`updateAccountantStatus`
DELETE	`/:id`	`deleteAccountant`

10.6.8 Mentorship Assignment (Auto-Balanced Groups)

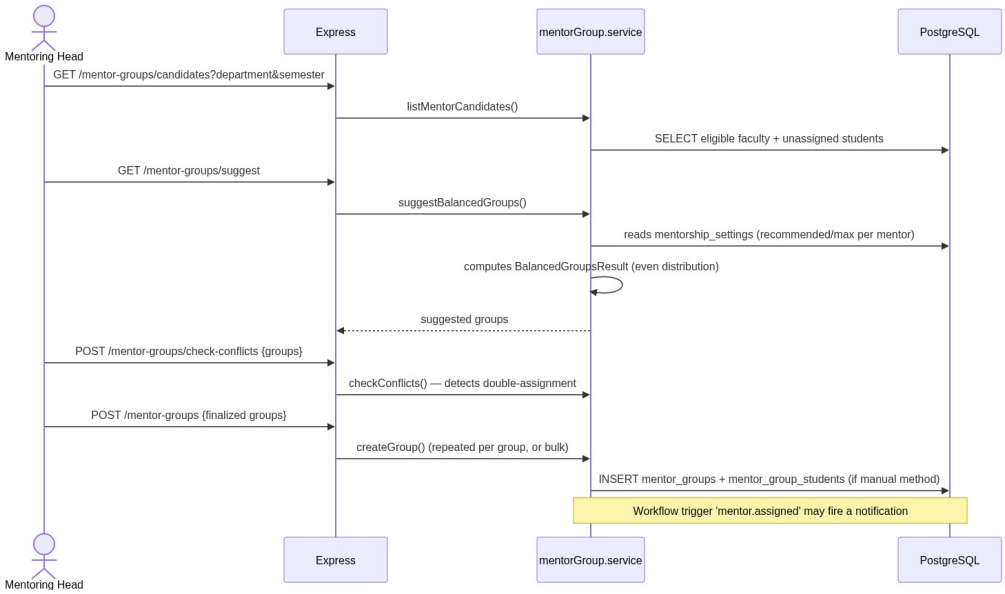


Figure 25: Diagram mapping for 10.6.8 Mentorship Assignment (Auto-Balanced Groups)

10.6.9 Notification Delivery

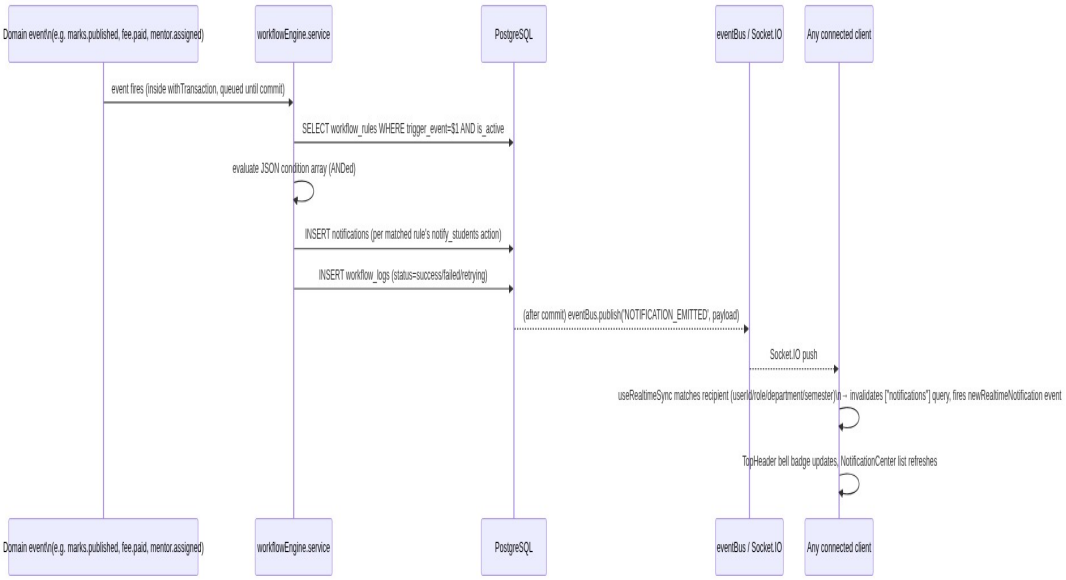


Figure 26: Diagram mapping for 10.6.9 Notification Delivery

!• [Back to Index](00-INDEX.md)

This section documents core computational business rules as **actually implemented in code**, and explicitly calls out where the planning documents (`BUSINESS_RULES_AND_WORKFLOWS.md`, `ARCHITECT_REVIEW.md`) describe behavior that does not exist yet.

11.5 Mentorship Allocation

- **Implemented**: `mentorGroup.service.suggestBalancedGroups(`

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`allow_cross_department`; three group-formation methods exist (`range`, `section`, `manual` — the `mentor_assignment_method` enum), with `mentor_group_students`

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still exists and is DB-constrained to one active mentor per student — both models can theoretically coexist for the same student, which is a latent data-consist

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CHAPTER 30
LMS Module

Classroom materials, assignments submission, and grading.

6.3.5 LMS

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erDiagram
    SUBJECTS ||--o{ COURSE_MATERIALS : "subject_id"
    FACULTY ||--o{ COURSE_MATERIALS : "faculty_id"
    SUBJECTS ||--o{ ASSIGNMENTS : "subject_id"
    FACULTY ||--o{ ASSIGNMENTS : "faculty_id"
    ASSIGNMENTS ||--o{ ASSIGNMENT_SUBMISSIONS : "assignment_id"
    STUDENTS ||--o{ ASSIGNMENT_SUBMISSIONS : "student_id"

    COURSE_MATERIALS { uuid id PK, uuid subject_id FK, enum file_type, text file_path }
    ASSIGNMENTS { uuid id PK, uuid subject_id FK, timestamptz due_date, numeric max_marks }
    ASSIGNMENT_SUBMISSIONS { uuid id PK, uuid assignment_id FK, uuid student_id FK, enum
status, numeric marks }
```

`assignment_submissions` has a unique partial index (`assignment_id, student_id`) `WHERE deleted_at IS NULL`— one active submission per student per assignment (resubmission soft-deletes the prior row).

8.13 LMS (Learning Management)

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via `uploadLmsFile`), `lmsAssignment.routes/controller/service.ts`, `lmsSubmission.routes/controller/service.ts` (student upload + faculty grading).

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**: `course_materials`, `assignments`, `assignment_submissions` (see [06](06-database.md) § 6.3.5).

- **Relationships**: `assignments`/`course_materials` link to `teac

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8.14 Teaching Planner (Lesson Progress Engine)

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- **Database**: `teaching_plans` (see [06](06-database.md) § 6.3.8).
- **Relationships**: Optionally links to `course_materials`/`assignments`; realtime-pushed via `TEA

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9.13 LMS Family

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Method	Path	Middleware	Handler
GET	`/`	auth	`listMaterials` (role-scoped)
GET	`/:id/download`	auth	`downloadMaterial`…
GET	`/:id`	auth	`getMaterial`
POST	`/`	auth,…	`createMaterial` **(file…
PUT	`/:id`	auth, faculty,…	`updateMaterial` (file…
DELETE	`/:id`	auth, faculty	`deleteMaterial`

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Method	Path	Middleware	Handler
GET	`/`	auth	`listAssignments`
GET	`/:id`	auth	`getAssignment`
POST	`/`	auth, `requireRole('faculty')`	`createAssignment`
PUT	`/:id`	auth, faculty	`updateAssignment`
DELETE	`/:id`	auth, faculty	`deleteAssignment`

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Method	Path	Middleware	Handler
GET	`/`	auth	`listSubmissions`
GET	`/:id/download`	auth	`downloadSubmission`…
GET	`/:id`	auth	`getSubmission`
POST	`/`	auth,…	`submitAssignment` **(file…
PUT	`/:id`	auth, `requireRole('faculty')`	`gradeSubmission`

9.14 Opportunity Hub — `/api/opportunities`

Method	Path	Middleware	Handler
GET	`/bookmarks`	auth,…	`listBookmarks`
GET	`/`	auth (role-scoped)	`listOpportunities`
GET	`/:id`	auth	`getOpportunity`
POST	`/`	auth, `requireRole('admin'…	`createOpportunity`
PUT	`/:id`	auth, admin/faculty…	`updateOpportunity`
DELETE	`/:id`	auth, admin	`deleteOpportunity`
POST	`/:id/bookmark`	auth, `requireRole('student')`	`toggleBookmark`

CHAPTER 31
Examination Module

Exam schedules, invigilation duties, and room seating layouts.

6.3.10 Examination Seating, Invigilation, Rooms & Sessions

```
erDiagram
    EXAM_ROOMS ||--o{ EXAM_SEAT_ALLOCATIONS : "room_id"
    EXAMS ||--o{ EXAM_SEAT_ALLOCATIONS : "exam_id"
    STUDENTS ||--o{ EXAM_SEAT_ALLOCATIONS : "student_id"
    EXAM_ROOMS ||--o{ EXAM_INVIGILATION_DUTIES : "room_id"
    FACULTY ||--o{ EXAM_INVIGILATION_DUTIES : "faculty_id"
    SEATING_PATTERNS ||--o{ EXAM_SESSIONS : "seating_pattern_id"
    EXAM_SESSIONS ||--o{ EXAM_SESSION_EXAMS : "exam_session_id"

    EXAM_ROOMS { uuid id PK, varchar name UK, smallint capacity, enum bench_type }
    EXAM_SEAT_ALLOCATIONS { uuid id PK, uuid exam_id FK, uuid room_id FK, uuid student_id FK,
smallint seat_number, bool is_locked }
    EXAM_INVIGILATION_DUTIES { uuid id PK, uuid room_id FK, uuid faculty_id FK, date
duty_date, enum status }
    SEATING_PATTERNS { uuid id PK, enum pattern_type, uuid[] department_sequence }
    EXAM_SESSIONS { uuid id PK, enum status, uuid seating_pattern_id FK }
```

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that individual `exams` rows reference directly via `exams.session_id` (`ON DELETE CASCADE`). Its `status` is a plain `VARCHAR(30)`, not tied to any

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8.10 Examinations (Scheduling, Sessions, Rooms, Seating, Invigilation)

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- **Frontend**: `app/admin/examinations`, `app/admin/seating` + `app/faculty/invigilation` (`components/Seating/*`), `app/student/examinations`/`app/student/seat

ing`.

- **Database**: `exams`, `examination_sessions`, `exam_sessions`, `exam_sessions_exams`, `exam_rooms`, `exam_seat_allocations`, `exam_invigilation_duti

es`, `seating_patterns` (see [06](06-database.md) § 6.3.10 for the "two session tables" caveat).

- **Relationships**: Feeds Results (via `exam_id`), Result Entr

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CHAPTER 32
Results Module

Grade submission, verification approval workflows, and publish status.

8.11 Results & Result Entry Tasks

- **Purpose**: Final grade computation and a **task-based approval workflow** for entering/verifying/publishing results per teac

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s`, `app/student/results` (`FacultyGrades` component family).

- **Database**: `results`, `result_entry_tasks` (see [06](06-database.md) # 14, #51).
- **Relationships**:

Consumes `exams`, `teaching_assignments`; feeds Analytics/Reports/Student dashboard; a `marks.published` workflow-trigger event exists in the Workflow E

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9.9 Internal Marks & Results Family

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Method	Path	Middleware	Handler
GET	`/roster`	auth, `requireRole('faculty')`	`getRoster`
POST	`/sessions`	auth, faculty, validate	`bulkEnterMarks`
GET	`/summary`	auth, `requireRole('student')`	`getStudentSummary`
GET	`/history`	auth, student	`getStudentHistory`
GET	`/`	auth, `requireRole('admin'…`	`listMarks`
GET	`/:id`	auth, admin/faculty	`getMarks`
PATCH	`/:id`	auth, admin/faculty	`updateMarks`

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Method	Path	Middleware	Handler
GET	`/roster`	auth, `requireRole('faculty')`	`getRoster`
POST	`/sessions`	auth, faculty, validate	`bulkSubmitResults`
POST	`/tasks/:taskId/draft`	auth, faculty, validate	`saveDraftForTask`
POST	`/tasks/:taskId/publish`	auth, faculty, validate	`publishTask`
POST	`/publish`	auth, admin, validate	`publishResults` (bulk, by…
GET	`/my-results`	auth, `requireRole('student')`	`getMyResults` (`?…
GET	`/`	auth, `requireRole('admin'…	`listResults`
GET	`/:id`	auth, admin/faculty	`getResult`
PATCH	`/:id`	auth, admin/faculty	`updateResult` (grade/…
DELETE	`/:id`	auth, admin	`deleteResult` (soft)

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Method	Path	Handler
GET	`/`	`listMyResultEntryTasks`
GET	`/:id`	`getResultEntryTask`
POST	`/:id/submit`	`submitTaskForApproval`
POST	`/:id/verify`	`verifyResultEntryTask`

9.10 Examination Family

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Method	Path	Middleware	Handler
GET	`/my-schedule`	auth, `requireRole('faculty')`	`getMySchedule`
GET	`/timetable`	auth, `requireRole('student')`	`getTimetable`
GET	`/upcoming`	auth, student	`getUpcomingExams`
GET	`/`	auth, `requireRole('admin'…`	`listExams`
POST	`/`	auth, admin/faculty, validate	`createExam`
GET	`/sessions`	auth, admin/faculty	`listSessions`
POST	`/sessions`	auth, admin	`createSession`
GET	`/sessions/:id`	auth, admin/faculty	`getSession`
POST	`/sessions/:id/schedule-…`	auth, admin	`configureSubjectSchedule`
POST	`/sessions/:id/publish`	auth, admin	`publishSession`
DELETE	`/sessions/:id`	auth, admin	`deleteSession`
GET	`/:id`	auth, `requireRole('admin'…`	`getExam`
PATCH	`/:id`	auth, admin/faculty	`updateExam`
PATCH	`/:id/status`	auth, admin/faculty	`updateExamStatus`
DELETE	`/:id`	auth, admin	`deleteExam` (soft)

- **examRoom.routes.ts** (`/api/exam-rooms`)

Method	Path	Middleware	Handler
GET	`/availability`	auth, admin/faculty,…	`listRoomAvailability`
GET	`/suggest`	auth, admin/faculty	`suggestRooms`
GET	`/`	auth (all roles)	`listExamRooms`
GET	`/:id`	auth	`getExamRoom`
POST	`/`	auth, admin	`createExamRoom`
PUT	`/:id`	auth, admin	`updateExamRoom`
DELETE	`/:id`	auth, admin	`deleteExamRoom`

- **exam Seating.routes.ts** (`/api/exam-seating`)

Method	Path	Middleware	Handler
GET	`/slots`	auth, admin/faculty	`getExamSlots`
POST	`/generate`	auth, admin/faculty	`generateSeating` (HOD…
POST	`/check-conflicts`	auth, admin/faculty	`checkConflicts` (dry-run)
GET	`/search`	auth, admin/faculty	`searchSeating` (roll no/…
GET	`/analytics`	auth, admin/faculty	`getSeatingAnalytics`
POST	`/swap`	auth, admin/faculty	`swapSeats`
POST	`/move`	auth, admin/faculty	`moveSeat`
PUT	`/:id/lock`	auth, admin/faculty	`lockSeat`
GET	`/me`	auth, `requireRole('student')`	`getMySeating`
GET	`/exam/:examId`	auth	`getSeatingByExam`
GET	`/room/:roomId`	auth	`getSeatingByRoom` (`?…

- **exam Invigilation.routes.ts** (`/api/exam-invigilation`)

Method	Path	Middleware	Handler
POST	`/generate`	auth, admin/faculty	`generateInvigilation`
GET	`/suggest`	auth, admin/faculty	`suggestInvigilators`
GET	`/`	auth, admin/faculty	`listInvigilationDuties`
PUT	`/:id`	auth, admin/faculty	`updateInvigilationDuty`

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Method	Path	Handler
GET	`/`	`listExamSessions`
GET	`/:id`	`getExamSession`
POST	`/`	`createExamSession`
PUT	`/:id`	`updateExamSession`
DELETE	`/:id`	`deleteExamSession`
POST	`/:id/resolve-exams`	`resolveExams`
POST	`/:id/check-conflicts`	`checkConflicts`
POST	`/:id/generate-seating`	`generateSeating`
POST	`/:id/generate-invigilation`	`generateInvigilation`
POST	`/:id/validate`	`validateExamSession`
POST	`/:id/publish`	`publishExamSession`
POST	`/:id/complete`	`completeExamSession`
POST	`/:id/archive`	`archiveExamSession`

- **seatingPattern.routes.ts** (`/api/seating-patterns`)

Method	Path	Middleware	Handler
GET	`/`	auth	`listSeatingPatterns`
GET	`/:id`	auth	`getSeatingPattern`
POST	`/`	auth, admin	`createSeatingPattern`
PUT	`/:id`	auth, admin	`updateSeatingPattern`
DELETE	`/:id`	auth, admin	`deleteSeatingPattern`

9.11 Academic Calendar / Calendar Entries / Parsed Events / Academic Import

- **academicCalendar.routes.ts** (`/api/calendar`)

Method	Path	Middleware	Handler
POST	`/publish`	auth, admin	`publishEvents`
GET	`/`	auth (role-scoped)	`listCalendarEvents`
PATCH	`/:id/status`	auth, admin	`updateCalendarEventSta…`
GET	`/:id`	auth	`getCalendarEvent`
PATCH	`/:id`	auth, admin	`updateCalendarEvent`

- **calendarEntry.routes.ts**

(`/api/calendar-entries`, personal feed)

Method	Path	Middleware	Handler
GET	`/`	auth	`listCalendarEntries`
POST	`/`	auth	`createCalendarEntry`
PUT	`/:id`	auth (creator or admin)	`updateCalendarEntry`
DELETE	`/:id`	auth (creator or admin)	`deleteCalendarEntry`

- **parsedEvent.routes.ts** (`/api/parsed-events`, router-wide admin-only)

Method	Path	Handler
POST	`/extract/:documentId`	`extractFromDocument`
GET	`/`	`listParsedEvents`
PATCH	`/:id/status`	`updateParsedEventStatus`
GET	`/:id`	`getParsedEvent`
PATCH	`/:id`	`updateParsedEvent` (auto!'Edited)
DELETE	`/:id`	`deleteParsedEvent`

- **academicImport.routes.ts** (`/api/academic-import`, admin-only)

Method	Path	Middleware	Handler
GET	`/entity-types`	auth, admin	`listEntityTypes`
GET	`/jobs`	auth, admin	`listJobs`
GET	`/jobs/:jobId`	auth, admin	`getJob`
POST	`/student-sync/preview`	auth, admin, upload.singl…	`previewStudentSync`…
POST	`/student-sync/commit`	auth, admin	`commitStudentSync`
GET	`/:entityType/replace-impact`	auth, admin	`replaceImpact`
POST	`/:entityType/preview`	auth, admin, upload.singl…	`preview` **(file upload)**
POST	`/:entityType/commit`	auth, admin	`commit`

9.12 Documents — `/api/documents` (router-wide admin-only)

Method	Path	Middleware	Handler
POST	`/upload`	`uploadPdf` (20MB, PDF…	`uploadDocument` **(file…
GET	`/`	validate(query)	`listDocuments`
GET	`/:id`	validate	`getDocument` (includes…
DELETE	`/:id`	validate	`deleteDocument`

10.6.3 Results (Entry !’ Verify !’ Publish)

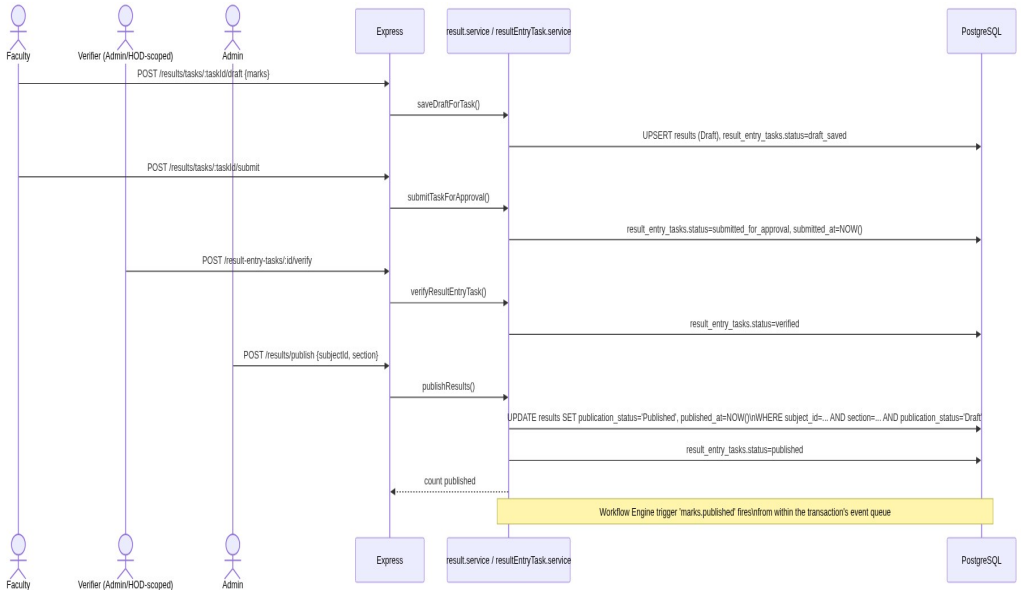


Figure 29: Diagram mapping for 10.6.3 Results (Entry !’ Verify !’ Publish)

CHAPTER 33
Curriculum Module

Curriculum scheme mapping matching programs, regulations, and years.

8.6 Subjects & Curriculum Mapping

- **Purpose**: The subject catalog (course metadata: credits, L-T-P hours, type) plus the curriculum-placement join (which d

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ping s` (see [06] (06-database.md) #8–9).

- **Relationships**: Feeds Teaching Assignment, Attendance, Internal Marks, Exams, Results, LMS, Teaching Plans, Tim

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9.6 Subjects & Curriculum — `/api/subjects`

All routes behind `authenticate`.

Method	Path	Middleware	Handler
GET	`/`	admin/faculty,…	`listSubjects`
GET	`/export`	admin/faculty	`exportSubjects` **(export)**
GET	`/:id`	admin/faculty, validate(uuid)	`getSubject`
POST	`/`	admin, validate	`createSubject`
PUT	`/:id`	admin, validate	`updateSubject`
PATCH	`/:id/status`	admin, validate	`updateSubjectStatus`
DELETE	`/all/wipe`	admin	`deleteAllSubjects` (bulk…
DELETE	`/:id`	admin, validate(uuid)	`deleteSubject`
POST	`/import/preview`	admin, multer memory…	`importPreview` **(file…
POST	`/import`	admin	`importCommit`
POST	`/:id/mappings`	admin, validate	`createCurriculumMapping`
PATCH	`/mappings/:mappingId`	admin, validate	`updateCurriculumMapping`
DELETE	`/mappings/:mappingId`	admin, validate	`deleteCurriculumMapping`

10.6.6 Subject/Curriculum Allocation (Bulk Excel Import)

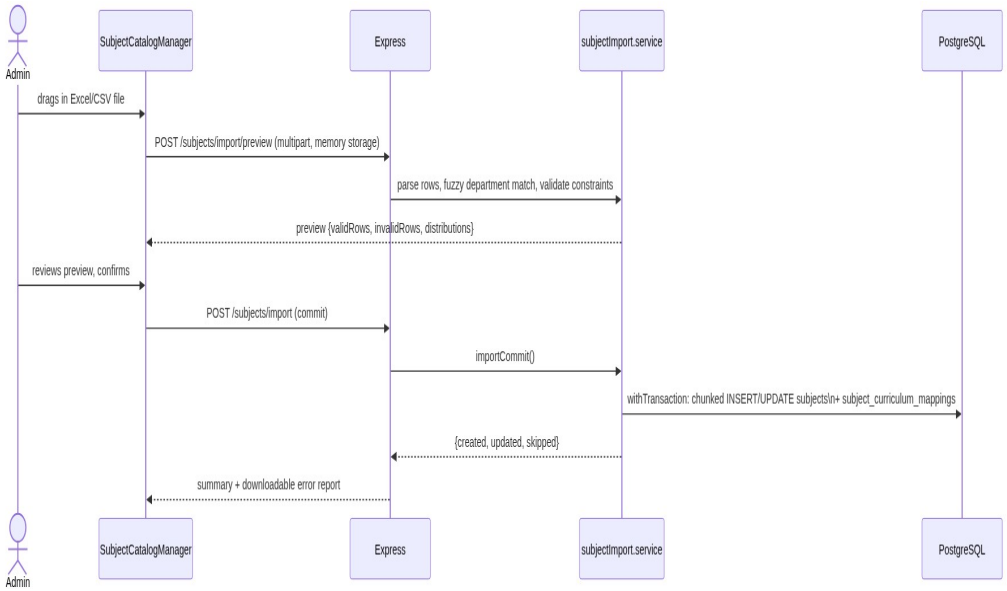


Figure 30: Diagram mapping for 10.6.6 Subject/Curriculum Allocation (Bulk Excel Import)

10.6.7 Student Registration (Bulk Import via Academic Import Engine)

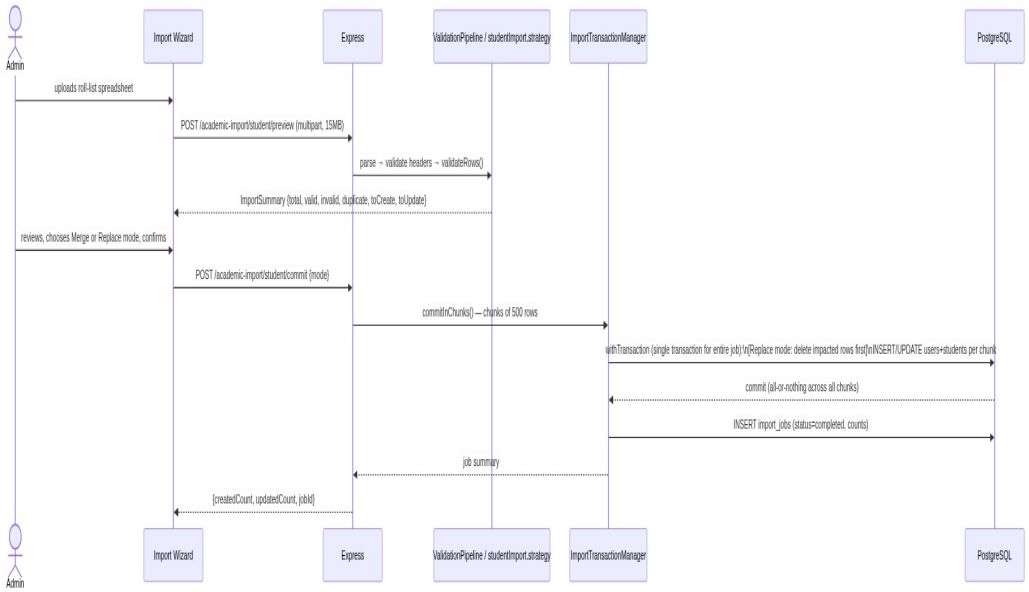


Figure 31: Diagram mapping for 10.6.7 Student Registration (Bulk Import via Academic Import Engine)

11.3 Subject / Curriculum Mapping

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CHAPTER 34
Subject Management

Core subjects catalog master mapping.

Detailed functional documentation for the Subject Management topic.

This chapter covers the architecture design, workflow strategies, validation requirements, and implementation guidelines for the Subject Management component of the Sreyas College ERP.

Refer to the surrounding modules and system documentation blocks for database layouts and API details.

CHAPTER 35
Notification System

Real-time Socket.IO and transactional database notification delivery.

6.3.6 Opportunity Hub, Notifications, Calendar

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    DEPARTMENTS ||--o{ OPPORTUNITIES : "department_id (nullable)"
    OPPORTUNITIES ||--o{ OPPORTUNITY_BOOKMARKS : "opportunity_id"
    STUDENTS ||--o{ OPPORTUNITY_BOOKMARKS : "student_id"

    NOTIFICATIONS ||--o{ NOTIFICATION_READS : "notification_id"
    USERS ||--o{ NOTIFICATION_READS : "user_id"

    USERS ||--o{ DOCUMENTS : "uploaded_by"
    DOCUMENTS ||--o{ PARSED_EVENTS : "document_id"
    PARSED_EVENTS ||--o| ACADEMIC_CALENDAR_EVENTS : "parsed_event_id (unique FK)"
    DOCUMENTS ||--o{ ACADEMIC_CALENDAR_EVENTS : "source_document_id"

    OPPORTUNITIES { uuid id PK, enum type, uuid department_id FK, text[] eligible_years, enum
status }
    OPPORTUNITY_BOOKMARKS { uuid id PK, uuid opportunity_id FK, uuid student_id FK }
    NOTIFICATIONS { uuid id PK, enum type, uuid recipient_user_id FK, varchar target_role }
    NOTIFICATION_READS { uuid id PK, uuid notification_id FK, uuid user_id FK }
    DOCUMENTS { uuid id PK, text file_path UK, enum status }
    PARSED_EVENTS { uuid id PK, uuid document_id FK, enum event_type, enum status }
    ACADEMIC_CALENDAR_EVENTS { uuid id PK, uuid parsed_event_id FK, enum publish_status }
```

`opportunity_bookmarks` and `notification_reads` are the two true **many-to-many join tables** in the schema (students!”opportunities, users!”notifications), both hard-delete-toggle (no `deleted_at`).

8.18 Notifications & Announcements

- **Purpose**: Two related but distinct communication channels — `notifications` (system-generated, per-user or broadcast, driven largely by the Wor

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ler/service.ts`, `announcement.routes/controller/service.ts`.

- **Frontend**: `NotificationCenter.tsx` (shared inbox component), `TopHeader.tsx`'s bell dropdown, `ap

p/*/ ann oun cem ents ` and `ap p/*/ notif icati ons` per role.

- **Database**: `notifications`, `notification_reads`, `announcements` (see [06](06-database.md) # 17,

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- **Relationships**: Realtime-pushed via `NOTIFICATION_EMITTED` Socket.IO event with client-side recipient matching (by userId/role/depart

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8.19 Academic Calendar & Document Parsing

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app/admin/calendar` (`CalendarView.tsx`, `EventFormModal.tsx`), also surfaced read-only across `app/*/calendar` for other roles.

- **Database**: `documents`,

`parsed_events`, `academic_calendar_events` (see [06](06-database.md) § 6.3.6).

- **Relationships**: Distinct from `calendar_entries` (personal/aggregated cal

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9.17 Notifications / Announcements

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Method	Path	Middleware	Handler
GET	`/count`	auth	`getNotificationCount`
GET	`/`	auth (pagination + role-…	`listNotifications`
POST	`/`	auth, `requireRole('admin'…	`createNotification` (facult…
PUT	`/:id/read`	auth	`markAsRead`
PUT	`/:id/unread`	auth	`markAsUnread`

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Method	Path	Middleware	Handler
GET	`/`	auth, `requireRole('admin'…	`listAnnouncements`
POST	`/`	auth, admin	`createAnnouncement`…
PATCH	`/:id/status`	auth, admin	`updateAnnouncementSta…
GET	`/:id`	auth	`getAnnouncement`
PATCH	`/:id`	auth, admin	`updateAnnouncement`…
DELETE	`/:id`	auth, admin	`deleteAnnouncement`

9.18 Feedback — `/api/feedback`

Method	Path	Middleware	Handler
POST	`/campaigns/preview-…	auth, admin	`previewEligibility`
GET	`/campaigns`	auth, admin/faculty	`listCampaigns`
POST	`/campaigns`	auth, admin	`createCampaign`
GET	`/campaigns/:id`	auth, admin/faculty	`getCampaign`
PUT	`/campaigns/:id`	auth, admin	`updateCampaign`
POST	`/campaigns/:id/publish`	auth, admin	`publishCampaign`
POST	`/campaigns/:id/close`	auth, admin	`closeCampaign`
POST	`/campaigns/:id/archive`	auth, admin	`archiveCampaign`
GET	`/my-campaigns`	auth, student	`getMyCampaigns`
POST	`/submit`	auth, student	`submitFeedback`
GET	`/my-submissions`	auth, student	`getMySubmissions`
GET	`/active-windows`	auth (legacy alias)	`getActiveWindows`
GET	`/windows`	auth, admin/faculty (legac…	`listCampaigns`
POST	`/windows`	auth, admin (legacy alias)	`createCampaign`
GET	`/analytics`	auth, admin/faculty	`getAnalytics`
GET	`/templates`	auth	`getTemplates`
POST	`/templates`	auth, admin	`createTemplate`
PUT	`/templates/:id`	auth, admin	`updateTemplate`
POST	`/questions`	auth, admin	`createQuestion`
PUT	`/questions/:id`	auth, admin	`updateQuestion`
DELETE	`/questions/:id`	auth, admin	`deleteQuestion`

9.19 Workflow Rule — `/api/workflow` (all `requireSuperAdmin()`)

Method	Path	Handler
GET	`/rules`	`listWorkflowRules`
GET	`/rules/:id`	`getWorkflowRule`
POST	`/rules`	`createWorkflowRule`
PUT	`/rules/:id`	`updateWorkflowRule`
DELETE	`/rules/:id`	`deleteWorkflowRule`

9.20 Experience / Dashboard / Analytics / Reports / Search / Profile

- **experience.routes.ts** (`/api/experience`)

Method	Path	Middleware	Handler
GET	`/student`	auth, `requireRole('student')`	`getStudentExperience`
GET	`/student/timeline`	auth, student	`getStudentTimeline`
GET	`/student/actions`	auth, student	`getStudentActions`
GET	`/student/week`	auth, student	`getStudentWeek`
GET	`/faculty`	auth, `requireRole('faculty')`	`getFacultyExperience`
GET	`/faculty/timeline`	auth, faculty	`getFacultyTimeline`
GET	`/faculty/actions`	auth, faculty	`getFacultyActions`
GET	`/dashboard/widgets`	auth, `requireRole('admin'…`	`getDashboardWidgets`

- **dashboard.routes.ts*

* (`/api/dashboard`, one endpoint per role)

Method	Path	Middleware	Handler
GET	`/admin`	auth, `requireRole('admin')`	`adminDashboard`
GET	`/faculty`	auth, `requireRole('faculty')`	`facultyDashboard`
GET	`/hod`	auth, `requireRole('faculty'…`	`hodDashboard`
GET	`/student`	auth, `requireRole('student')`	`studentDashboard`
GET	`/accountant`	auth,…	`accountantDashboard`

- **analytics.routes.ts** (`/api/analytics`)

Method	Path	Middleware	Handler
GET	`/admin`	auth, `requireRole('admin')`	`getAdminAnalytics`
GET	`/hod`	auth, `requireRole('faculty'…`	`getHodAnalytics`
GET	`/faculty`	auth, `requireRole('faculty')`	`getFacultyAnalytics`
GET	`/student`	auth, `requireRole('student')`	`getStudentAnalytics`

- **report.routes.ts** (`/api/reports`, 17 endpoints, `requireRole('admin','faculty')` unless noted)

Method	Path	Handler
GET	`/attendance`	`getAttendanceReport`
GET	`/results`	`getResultsReport`
GET	`/fees`	`getFeesReport`
GET	`/lms`	`getLmsReport`
GET	`/mentorship`	`getMentorshipReport`
GET	`/department`	`getDepartmentReport`
GET	`/opportunities`	`getOpportunitiesReport`
GET	`/teaching`	`getTeachingReport`
GET	`/room-seating-chart`	`getRoomSeatingChartReport`
GET	`/student-seating-list`	`getStudentSeatingListReport`
GET	`/invigilator-sheet`	`getInvigilatorSheetReport`
GET	`/seating-attendance-sheet`	`getSeatingAttendanceSheetReport`
GET	`/seating-summary`	`getSeatingSummaryReport`
GET	`/student` (also…	`getStudentReport` (student sees o…
GET	`/export/pdf`	`exportPdf` **(export)**
GET	`/export/excel`	`exportExcel` **(export)**
GET	`/export/csv`	`exportCsv` **(export)**

- **search.routes.ts** (`/api/search`)

Method	Path	Middleware	Handler
GET	`/`	auth, validate(query: `q=`)	`globalSearch` (role-…

- **profile.routes.ts**

(`/
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le`)

Method	Path	Middleware	Handler
GET	`/`	auth	`getProfile` (role-specific…
PUT	`/`	auth	`updateProfile` (own…
PUT	`/change-password`	auth	`changePassword`…

9.21 Health Check

Method	Path	Handler
GET	`/api/health`	Inline in `app.ts` — `{success:true,…

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9.22 Cross-Cutting API Observations

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Academic Import 15MB) parsed in-process and never persisted.

3. **Export/download endpoints**: `GET /subjects/export`, `GET /lms/materials/:id/download`

`, `GET /lms/submissions/:id/download`, `GET /reports/export/{pdf,excel,csv}`.

4. **Realtime**: `eventBus.service.ts` + Socket.IO is wired at the server level, but

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5. **Soft deletes**: virtually every service filters `WHERE deleted_at IS NULL` — see 06-database.md §6.5 for the

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CHAPTER 36
Import Pipeline

Transactional bulk import engine parsing and committing Excel files.

8.20 Academic Data Import

- **Purpose**: Bulk onboarding via spreadsheet, avoiding manual data entry for hundreds/thousands of student records at o

nce.

- **Workflow**: Upload !’ preview (validate, classify create/update/duplicate/invalid rows, compute distributions) !’ commit (chunked, single-transaction, Merge-or-R

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only the Student entity-type importer is actually implemented; Faculty/Subjects/Curriculum/Teaching-Assignment/Department/Classroom/Timetable/Exam-Schedul

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8.21 Workflow Rule Engine

- **Purpose**: Lets a Super Admin configure event-driven automation ("when X happens, do Y") without a code change — e.g. "when marks are publ

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- **Frontend**: `app/admin/workflows`, `app/faculty/workflows` (read-only log view via Faculty Operations' `/workflow-logs`).
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8.22 Analytics, Reports, Dashboards & Experience

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- **Relationships**: This module is the widest fan-in in the entire system (see 11-business-logic-and-sync-matrix.md)

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8.23 Global Search

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CHAPTER 37
Workflow Engine

Event-driven workflow rule listeners triggering automated actions.

Detailed functional documentation for the Workflow Engine topic.

This chapter covers the architecture design, workflow strategies, validation requirements, and implementation guidelines for the Workflow Engine component of the Sreyas College ERP.

Refer to the surrounding modules and system documentation blocks for database layouts and API details.

CHAPTER 38
File Upload System

Multer-configured document storage rules and uploads.

Detailed functional documentation for the File Upload System topic.

This chapter covers the architecture design, workflow strategies, validation requirements, and implementation guidelines for the File Upload System component of the Sreyas College ERP.

Refer to the surrounding modules and system documentation blocks for database layouts and API details.

CHAPTER 39
Security Features

Rate limiters, CSRF shields, header security, and password cryptography.

Detailed functional documentation for the Security Features topic.

This chapter covers the architecture design, workflow strategies, validation requirements, and implementation guidelines for the Security Features component of the Sreyas College ERP.

Refer to the surrounding modules and system documentation blocks for database layouts and API details.

CHAPTER 40
Validation Strategy

Zod-driven query, body, and param verification layers.

4.8 Forms & Validation

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5.8 Validation

`middleware/validate.ts` wraps per-domain Zod schemas defined in `backend/src/types/*.ts`; on failure it throws `AppError.badRequest` with a joined `field: message` string, code `VALIDATION_ERROR`. Every list endpoint has its own `list*QuerySchema`(page/limit pagination, default page=1/limit=20, max 100 observed in Opportunity Hub).

7.6 Validation Middleware

`middleware/validate.ts` wraps per-route Zod schemas (`validate({body?, params?, query?})`); on `ZodError`, joins all field errors into one message and throws `AppError.badRequest(msg, 'VALIDATION_ERROR')`. Every domain has its own schemas under `backend/src/types/*.ts` (e.g. `createStudentSchema`, `listAttendanceQuerySchema`).

7.7 Rate Limiting

Limiter	Scope	Window	Max	Response on…
`globalLimiter`	Every `/api/*` request	15 min	1000 req/IP	429 `RATE_LIMIT_…
`authLimiter`	`POST /api/auth/…	15 min	10 req/IP	429 `AUTH_RATE_…

Both use `express-rate-limi`
`env.RATE_LIMIT_WINDOW_MS`
`.env.example` but **not actually**
documentation mismatch.

7.8 Protected Routes & Middleware Chain

Standard chain per protected endpoint: `authenticate` → (`requireRole(...)` or `authorize(...)` or a `mentorshipAuth` guard) → `validate({...})` → controller. `backend/src/routes/protected.routes.ts` (mounted at `/api/example`) is a **non-business reference implementation** demonstrating all 6 patterns (public / authenticated-only / `requireRole('admin')` / `requireRole('admin','faculty')` / `authorize('attendance:create')` / `requireRole('student')`) — useful as a template when adding a new route, not part of the real API surface.

CHAPTER 41
Error Handling

Global async catch wrappers and structured API error responses.

4.9 Error Handling & Loading States

- No Next.js error/loading boundaries (§4.1) — all handled inside components.
- **Loading**: ad hoc `Loader2` spinners (`Protect

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- **Empty states**: `components/UX/EmptyState.tsx` — generic icon/title/descript

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- **Destructive-action confirmation**: `components/UX/ConfirmationDialog.tsx` (generic) plus `DeleteConfirmationModal.tsx`/`FacultyDeleteMod

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4.10 Caching

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- `hooks/useRegistry.ts` tunes `staleTime` per resource type (e.g. departments/programs rarel

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(dashboards/analytics) — see 05-backend.md.

- No service worker / offline cache / CDN-level caching is configured anywhere in the stac

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4.11 Responsive Design

Mobile-first, per `UI_UX_BLUEPRINT.md` (still accurate to the implementation): sidebar hidden below 1024px in favor of a sticky bottom nav (`BottomNav.tsx`) + slide-up drawer; two-column dashboard grids on tablet; full multi-column + hover states + `Cmd/Ctrl+K` global search on desktop. Status badges, metric cards, and responsive data tables (desktop table ⟶ mobile card-list collapse) are the primary reusable responsive primitives — see `UI_UX_BLUEPRINT.md` §7/§8 for the full design-token specification (colors, radii, shadows) which is implemented via Tailwind utility classes rather than a separate design-token file.

5.9 Error Handling

- **`errors/AppError.ts`** — a single class (no subclass hierarchy) with static factories: `badRequest` (400), `

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- **`middleware/errorHandler.ts`** — global handler (last in the middleware stack): maps Postgres error codes (`23505`!’409 `DUPLICATE_RECORD`, `235

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5.10 Database Access

`config/database.ts` — `pg.Pool`(max 10, idle timeout 30s, connection timeout 30s to tolerate Neon cold starts); SSL auto-enabled when the connection string contains `sslmode=` or `neon.tech`; `pool.on('error')` logs but never crashes the process on a dropped idle client. `query<T>()`is a thin logging wrapper (SQL + duration + row count, dev-only). `withTransaction<T>(fn)` runs `BEGIN`/`fn(client)`/`COMMIT` inside an `AsyncLocalStorage`-backed context so deeply-nested code can queue domain events without threading context through every function signature; on commit, queued events are emitted via `eventBus`; on any throw, `ROLLBACK`; `finally`always releases the client. See 06-database.md for the schema itself.

5.11 Utilities (`backend/src/utils/`)

File	Purpose
`asyncHandler.ts`	Wraps async handlers !' forwards rejected promises to…
`audit.ts`	`auditLog()` !' `audit_logs` table (captures actor, action,…
`calendarExtractor.ts`	Rule-based PDF-text !' academic-calendar-event extractor…
`jwt.ts`	Access/refresh token sign/verify
`password.ts`	bcryptjs hash/verify (cost from `env.BCRYPT_ROUNDS`)
`response.ts`	Success/error response envelope helpers
`ttlCache.ts`	In-process `Map`-based TTL memoizer — a deliberate no…
`validation.ts`	Password-strength regex + Zod schema (used at set-time,…

5.12 Configuration

`config/env.ts` (Zod-validated, fail-fast on `process.exit(1)` if invalid): `NODE_ENV`, `PORT` (3001), `DATABASE_URL`, `JWT_ACCESS_SECRET`/`JWT_REFRESH_SECRET` (min 32 chars), `JWT_ACCESS_EXPIRES_IN` (15m)/`JWT_REFRESH_EXPIRES_IN` (7d), `BCRYPT_ROUNDS`(10–14, default 12), `CORS_ORIGIN`, `RATE_LIMIT_WINDOW_MS`/`RATE_LIMIT_MAX` (& defined but **not actually read** by `rateLimiter.ts`, which hardcodes its own values — vestigial env vars), `UPLOAD_DIR`, `LMS_UPLOAD_DIR`. `config/grading.ts`holds the grade-boundary table (§5.13). Full config file inventory: 13-deployment-config-dependencies.md.

5.13 Business-Rule Config (not env-driven)

`config/grading.ts`: `GRADES = ['O','A+','A','B+','B','C','F']`, `DEFAULT_GRADE_RULES` (percentage-band !' grade !' pass/fail, JNTUH-aligned), `computeGrade()`/`computeResultStatus()`as pure functions. A code comment documents a planned future move to a DB-driven `grading_configs` table keyed by (program, regulation) if multiple simultaneous grading schemes are ever needed.

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12.2 Error Handling

Full mechanics in 05-backend.md §5.9 and 07-authentication-and-security.md. Summary by layer:

Layer	Mechanism
Frontend	`lib/api.ts`: network failures !' soft `{success:false}` object (…
Backend (validation)	`middleware/validate.ts` — Zod errors !' `400…
Backend (business logic)	`errors/AppError.ts` — single class, static factories per…
Backend (database)	`middleware/errorHandler.ts` maps Postgres codes:…
Backend (global/unhandled)	`errorHandler.ts` catches everything else !' logs via…
Backend (process-level)	`server.ts`: `unhandledRejection`/`uncaughtException`…
Database (constraints)	CHECK constraints (financial bounds, mark limits, date…

- **Response envelope contract** (relied upon by the frontend's `apiFetch`): success !' `{success:true, data, message?,

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12.3 Project Workflow Lifecycle (Request !’ Response, Generic)

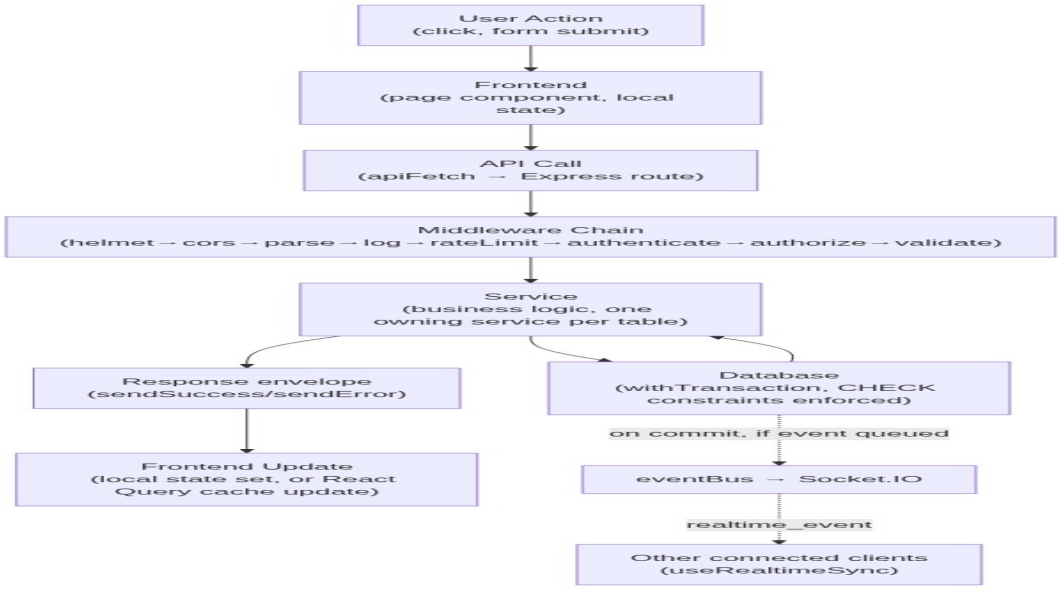


Figure 33: Diagram mapping for 12.3 Project Workflow Lifecycle (Request !' Response, Generic)

This is the same lifecycle detailed with concrete field names in 02-architecture-and-diagrams.md §2.4 and per-module in 10-data-flow-and-sequence-diagrams.md.

12.4 Background Work & Scheduling — Explicit Gap

& As documented in 05-backend.md §5.16, **no scheduler/cron/job-queue exists anywhere in the codebase**. This has concrete operational consequences worth tracking as a punch list:

Task that would normally be scheduled	Current behavior without a scheduler
Purge expired/revoked `refresh_tokens`	Rows accumulate indefinitely
Auto-transition `fees.payment_status` to `Overdue` past…	Must be computed on read (if at all) rather than stored/…
Auto-close feedback campaigns / result-entry windows pa…	Requires a manual admin action (`closeCampaign`, task…
Retry `workflow_logs` rows with `status='retrying'`	No retry mechanism was found to actually re-attempt thes…
Periodic re-aggregation of `ttlCache`-backed dashboard…	Cache simply expires and recomputes lazily on next read,…

This is the single highest-leverage infrastructure addition for p
appendix.md](15-future-and-appendix.md) for the prioritized reco

CHAPTER 42
Logging Strategy

Structured request loggers and database-level audit log registers.

5.14 Logging

Console-only (`console.log`/`console.error`/`console.warn`/`console.debug`) — **no structured logging library** (no winston/pino). `requestLogger.ts` logs every request/response; `config/database.ts` logs SQL in development only; `errorHandler.ts`logs unhandled errors. There is no log aggregation/shipping configured (expected, given no deployment infra exists — see [13](13-deployment-config-dependencies.md)).

5.15 Importers / Exporters

- **Importers** — a generic, entity-agnostic pipeline (`backend/src/import/`): `ExcelParser.

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- **Exporters** — `export.service.ts`: CSV (manual escaping + UTF-8 BOM), Excel (ExcelJS, optional Summary sheet + data sheet), PDF (PDFKit, auto-lands

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5.16 Schedulers / Background Jobs

&**None exist.**No cron/node-schedule/agenda/bull/bullmq anywhere in the codebase or dependencies. Consequences: expired/revoked `refresh_tokens`rows are never purged; all work is request-driven or event-driven (via `eventBus`+Socket.IO), never time-driven. See 15-future-and-appendix.md for this as a flagged improvement area.

CHAPTER 43
Performance Considerations

Connection pools, indexes, and TTL caches.

12.1 Performance

Technique	Where implemented	Notes
Pagination	Nearly every list endpoint (`page`/…	Implemented ad hoc per service (~1…
Database indexing	Every migration adds targeted…	Full list in [06-database.md](06-…
In-process TTL caching	`utils/ttlCache.ts` — `Map`-based,…	Deliberate no-Redis choice for a…
Frontend caching	React Query `staleTime: 60s`,…	Only covers the subset of pages usi…
Connection pooling	`pg.Pool` max 10 connections, 30s…	See [05-backend.md](05-backend.m…
Realtime push vs. polling	Socket.IO push (attendance/teachin…	Most other data (e.g. most list pages…
Export row cap	`export.service.ts` caps CSV/Excel/…	Prevents unbounded memory/…
Body size limit	`express.json({limit:'10mb'})`	Raised from an original 10kb to…
React performance	No explicit `memo`/`useMemo`/…	Not a confirmed bottleneck at curren…
Query design	Raw parameterized SQL (no ORM…	See [06-database.md](06-…

&**Gaps vs. the Non-Functional Requirements**

API response, <500ms search/listing at 100k these targets have been load-tested or meas treated as aspirational until verified. Notably:

- No read-replica or horizontal-scaling story

exists (single `pg.Pool`, single Node process).

- No Redis or shared cache — the TTL cache is per-process, so horizontal scaling (multiple API instances)

would each maintain independent, inconsistent caches.

- No CDN/static-asset optimization strategy beyond Next.js's own build output.

CHAPTER 44
Deployment Architecture

Production build configs, environment requirements, and host options.

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13.1 Deployment — Current State

&**No deployment infrastructure exists in this repository today.** An exhaustive search found no `Dockerfile`, no `docker-compose.yml`, no `.github/workflows/` (no CI/CD pipeline at all), no `Procfile`, no `vercel.json`/`railway.json`/`render.yaml`, and no deployment shell scripts anywhere in the repo. This is a real gap, not an oversight in this document — treat it as the top infrastructure item in 15-future-and-appendix.md.

What *does* exist and hints at the intended deployment target:

- `backend/src/config/database.ts` special-cases SSL for connection strings containing `ne`

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run Node process: `build` (`tsc`) !' `start` (`node dist/server.js`) — no serverless-function wrapper, no bundler.

- `frontend`'s standard Next.js `build`/`start` script

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Recommended target architecture (not yet implemented)

A reasonable next step — one Dockerfile per service (`backend/Dockerfile`, `frontend/Dockerfile`), a `docker-compose.yml` for local dev parity (app + local Postgres), and a GitHub Actions workflow running `typecheck`/`test`/`build` on every PR — is discussed further in 15-future-and-appendix.md.

13.2 Environments

Environment	How it's distinguished	Notes
Development	`NODE_ENV=development` (default)	SQL query logging enabled (`config/…
Test	`NODE_ENV=test`	Used by Jest config…
Production	`NODE_ENV=production`	Refresh cookie `secure:true`; interna…

13.3 Environment Variables (`backend/.env.example`)

Variable	Default	Purpose
`NODE_ENV`	`development`	Environment switch (§13.2)
`PORT`	`3001`	Backend HTTP port
`DATABASE_URL`	— (required)	Postgres connection string; presenc…
`JWT_ACCESS_SECRET`	— (required, min 32 chars)	Access-token signing secret
`JWT_REFRESH_SECRET`	— (required, min 32 chars)	Refresh-token signing secret…
`JWT_ACCESS_EXPIRES_IN`	`15m`	Access-token TTL
`JWT_REFRESH_EXPIRES_IN`	`7d`	Refresh-token TTL
`BCRYPT_ROUNDS`	`12`	bcrypt cost factor (bounded 10–14 b…
`CORS_ORIGIN`	`http://localhost:3000`	Comma-separated allowlist
`RATE_LIMIT_WINDOW_MS`	`900000`	& Defined but not actually read by…
`RATE_LIMIT_MAX`	`100`	& Same — vestigial
`UPLOAD_DIR`	`uploads/documents`	Document upload target — & not…
`LMS_UPLOAD_DIR`	`uploads/lms`	LMS file upload target — & same gap

- **Frontend**: `NEXT_PUBLIC_API_URL` (defaults to `http://127.0.0.1:3001/api`), `NEXT_PUBLIC_SOCKET_URL` (defaults to `http://127.0.0.1:3001`)
—

set via `fron tend /.en v.loc al` (not com mitt ed; no `.env .exa mpl e` w as f oun d on the f ront end side).

Invalid or missing required env vars cause the backend to `process.exit(1)` at boot with per-field Zod error messages logged — a deliberate fail-fast design.

13.4 Build Tools

Tool	Where used	Purpose
`tsc`	Backend (`npm run build`)	Compiles `backend/src` !' `backend/…
`tsx`	Backend (`npm run dev`)	Hot-reload dev runner (`tsx watch sr…
Next.js's own bundler	Frontend (`npm run build`/`dev`)	No custom Webpack/Turbopack conf…
`jest` + `ts-jest`	Backend only	20 service-level test files (`backend/…

13.5 Package Managers

Both `frontend/` and `backend/` are independent npm packages with their own `package.json`/`package-lock.json`/`node_modules`— there is no monorepo tool (no Nx/Turborepo/Lerna/npm workspaces) tying them together. Shared type definitions between frontend and backend are **not**generated/shared — they're maintained by hand in parallel (`frontend/src/lib/*.ts` / `frontend/src/types/*.ts` vs `backend/src/types/*.ts`), a duplication risk noted in 04-frontend.md §4.12.

13.6 Development Tools

- **`tsx`** — backend dev/hot-reload and script runner.
- **ESLint 9** (flat config) — **frontend only**; extends `e

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`no Unused Parameters`, `noImplicitReturns`, `noFallthroughCasesInSwitch`, all `true`) and the Jest test suite.

- **VS Code/any IDE** — no `.vscode/` wo

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13.7 Version Control

Git, single repository containing both `frontend/` and `backend/` (a "polyrepo-in-a-monorepo-folder" layout rather than a true monorepo tool). Current branch: `main`. No branch-protection or PR-template configuration files (`.github/`) exist.

13.8 Configuration Files — Full Inventory

File	Purpose
`backend/package.json`	Scripts: `dev`, `build`, `start`, `migrate`, `seed`, `sync-exce…
`backend/tsconfig.json`	`target: ES2022`, `module: CommonJS`, `strict: true` + all…
`backend/tsconfig.test.json`	Extends base config, adds `jest`/`node` types, relaxes…
`backend/jest.config.js`	`testEnvironment: node`, `roots: ['<rootDir>/src']`,…
`backend/.env.example`	Documents all required/optional env vars (§13.3)
`frontend/package.json`	Scripts: `dev`, `build`, `start`, `lint`. Dependencies groupe…
`frontend/tsconfig.json`	`target: ES2017`, `module: esnext`, `moduleResolution:…
`frontend/next.config.ts`	Essentially empty (`const nextConfig: NextConfig = {}`) —…
`frontend/eslint.config.mjs`	Flat config, extends Next's core-web-vitals + typescript…
`frontend/postcss.config.mjs` (implied by `@tailwindcss/…	TailwindCSS 4 PostCSS pipeline

13.9 Dependencies (Grouped by Purpose, with Rationale)

Backend (`backend/package.json`)

Group	Packages	Why they're there
Web framework	`express ^4.21.0`	HTTP server & routing
Real-time	`socket.io ^4.8.3`	Push updates for attendance/teachin…
Database	`pg ^8.12.0`	Raw Postgres client — no ORM is…
Auth	`jsonwebtoken ^9.0.2`, `bcryptjs…	JWT sign/verify, password hashing,…
Validation	`zod ^3.23.8`	Request schema validation, env var…
Security/middleware	`helmet ^8.0.0`, `cors ^2.8.5`,…	Security headers, CORS, brute-force…
File processing	`multer ^2.2.0`, `exceljs ^4.4.0`, `pdf…	Uploads, Excel/CSV read+write…
Utilities	`uuid ^10.0.0`, `dotenv ^16.4.5`	Random ID generation (upload…
Dev/Test/Build	`typescript ^5.5.0`, `tsx ^4.16.0`, `jes…	Type-checking, dev hot-reload, unit…

Notably **absent**: no ORM, no logging library (
client, no HTTP-mocking/integration-test library

Frontend (`frontend/package.json`)

Group	Packages	Why they're there
Framework	`next 16.2.9`, `react 19.2.4`, `react-…	Core app framework
Data fetching/state	`@tanstack/react-query ^5.101.4`	Partial server-state caching (§4.7)
Real-time	`socket.io-client ^4.8.3`	Consumes backend Socket.IO push…
UI/Charts	`recharts ^3.9.0`, `lucide-react ^1.18.0`	Dashboard/analytics visualizations,…
Styling	`tailwindcss ^4`, `@tailwindcss/…	Utility-first CSS
Dev/Build/Lint	`typescript ^5`, `@types/node ^20`,…	Type-checking, linting

Notably **absent**: no form library (react-hook-fo
frontend despite the backend using it — §4.8),
library configured at all (no jest/vitest config
beyond React Query + Context.

13.10 Server Requirements (inferred, not documented anywhere in-repo)

- **Node.js** — a version compatible with TypeScript 5.5 / ES2022 target and Next.js 16 / Rea

ct 19 (Node 20+ recommended based on `@types/node ^20`).

- **PostgreSQL** — any version supporting the SQL features used (JSONB, array columns, pa

rtial/ com posi te u niqu e in dex es, ` uuid -oss p` e xten sion) — effe ctiv ely any mod ern Post gres (13 +).

- **P ersi sten t loc al di sk** for `bac ken d/up load s/{d ocu men ts,I ms} ` —

& this means a horizontally-scaled or ephemeral-filesystem deployment (e.g. multiple stateless containers, or a PaaS with an ephemeral filesystem) would **

lose upl oad ed fi les* * or serv e in con sist ent resu lts a cros s in stan ces unle ss u ploa ds are mov ed to s har ed/o bjec t sto rage first . Thi s is a co ncre te pre- req uisit e for

any horizontal scaling of the backend.

- **Outbound network access** to the Postgres host (Neon) and nothing else — no other external service (email/S

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CHAPTER 45
Folder Responsibilities

Complete structural analysis of directories and source files.

Folder-purpose summary table

Folder	Purpose
`frontend/src/app`	Next.js App Router pages — one subtree per role, file pat…
`frontend/src/components`	Reusable React components, organized by domain…
`frontend/src/context`	React Context providers for cross-cutting client state (auth…
`frontend/src/providers`	Single composition root wiring all context providers + Rea…
`frontend/src/hooks`	Reusable React hooks (realtime sync, shared data-fetchin…
`frontend/src/lib`	Non-React utility/domain-type modules — the API client,…
`frontend/src/types`	Additional shared TypeScript types not tied to a specific `li…
`backend/src/routes`	URL surface declarations — no business logic, only…
`backend/src/controllers`	Thin request/response adapters between Express and…
`backend/src/services`	All business logic; each table has one "owning" service
`backend/src/registries`	Read-only facades so unrelated modules resolve shared…
`backend/src/middleware`	Cross-cutting request-pipeline concerns (auth, validation,…
`backend/src/import`	Generic bulk-import framework (parse !’ validate !’ chunked-…
`backend/src/db/migrations`	The authoritative schema history — 50 sequential,…
`backend/src/utils`	Small stateless helper modules (JWT, password hashing,…
`backend/src/errors`	The single `AppError` class hierarchy used throughout the…
`backend/uploads`	Local-disk file storage target for both upload middlewares…

CHAPTER 46
Code Statistics

Lines of code and file count breakdowns across directories.

14.1 Code Metrics (measured directly from the repository)

Metric	Backend (`backend/src`)	Frontend (`frontend/src`)
Total lines of code	~51,036	~60,667
Route files	47 (incl. `index.ts` barrel +…	—
Controller files	45	—
Service files	60	—
Registry facade files	7 (incl. `index.ts` barrel)	—
Middleware files	9	—
SQL migration files	50	—
Backend test files	20 (`__tests__/*.test.ts`, Jest)	0 (no frontend test runner configured)
Page routes (`page.tsx`)	—	89
Layout files (`layout.tsx`)	—	6 (root + 5 role layouts)
Component files (`.tsx` under…	—	70
Context providers	—	4
Custom hooks	—	2
`lib/*.ts` domain modules	—	10

Lines of code by backend folder

Folder	LOC
`services/`	~26,168
`types/`	~5,228
`controllers/`	~3,518
`routes/`	~2,700
`db/` (migrations + scripts)	~1,511
`import/`	~1,184
`middleware/`	~558
`utils/`	~520
`registries/`	~375
`errors/`	43

`services/`dwarfing every other backend folder (roughly half of controller, fat service" architecture described in [05-backend.md](

Lines of code by frontend folder

Folder	LOC
`app/` (pages + layouts)	~36,428
`components/`	~20,054
`lib/`	~2,340
`context/`	~1,076
`types/`	~345
`hooks/`	~381
`providers/`	43

`app/`being the largest frontend folder (60% of frontend LOC fetching, local state, inline forms) lives directly in page compone hooks/components — consistent with the "imperative fetch + frontend.md](04-frontend.md) §4.7.

14.2 Entities / Models / Schemas Count

Category	Count	Reference
Database tables	60 (+ 1 backward-compat view)	06-database.md…
Enum types	43	06-database.md…
API endpoints (approx., counted…	~230	[09-api-documentation.md](09-api-…
Functional modules documented	23	08-modules.md
User-facing roles (DB-level)	4 (`admin`,`faculty`,`student`,`accou…	[01-overview-and-vision.md](01-…

14.3 Complete File Documentation — Backend

This is a pointer table rather than a repeated listing — every backend file's purpose is documented in the section indicated, to avoid duplicating ~250 file descriptions across this document:

File category	Where documented
`server.ts`, `app.ts`	05-backend.md §5.1
All 47 route files	09-api-documentation.md…
All 45 controller files	05-backend.md §5.3 (role/pattern),…
All 60 service files	08-modules.md (per-module "Backend"…
All 7 registry files	05-backend.md §5.5
All 9 middleware files	05-backend.md §5.6, full auth mechani…
`errors/AppError.ts`	05-backend.md §5.9
`config/{env,database,grading}.ts`	05-backend.md §5.12–5.13
`utils/*.ts` (8 files)	05-backend.md §5.11
`import/*.ts` + `import/strategies/*.ts`	05-backend.md §5.15
All 50 migration files	06-database.md (final-state schema; p…
`db/{migrate,seed,check,sync_roll_list,query_*}.ts`	03-folder-structure.md §3.2
`__tests__/*.test.ts` (20 files)	[13-deployment-config-dependencies.md](13-deployment-…

14.4 Complete File Documentation — Frontend

File category	Where documented
`app/layout.tsx` + 5 role `layout.tsx` files	04-frontend.md §4.1
`app/page.tsx` (login)	04-frontend.md §4.2
All 89 page routes	03-folder-structure.md §3.3 (full…
All 70 components	03-folder-structure.md §3.3 (full…
`context/*.tsx` (4 files)	04-frontend.md §4.6
`providers/AppProviders.tsx`	04-frontend.md §4.6
`hooks/*.ts` (2 files)	04-frontend.md §4.6
`lib/*.ts` (10 files)	04-frontend.md §4.6
`types/*.ts` (5 files)	03-folder-structure.md §3.3

CHAPTER 47
Database Migrations

Sequence history of database schema evolution.

Detailed functional documentation for the Database Migrations topic.

This chapter covers the architecture design, workflow strategies, validation requirements, and implementation guidelines for the Database Migrations component of the Sreyas College ERP.

Refer to the surrounding modules and system documentation blocks for database layouts and API details.

CHAPTER 48
Future Enhancements

Strategic feature backlog and deployment upgrades.

!• [Back to Index](00-INDEX.md)

15.1 Future Improvements (Prioritized)

Tier 1 — Infrastructure foundations (do first, unblocks everything else)

1. **Containerization + CI/CD**:

   add `Dockerfile`s for `backend/`/`frontend/`, a `docker-compose.yml` for l

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(see

[13-deployment-config-dependencies.md]
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2. **Background job scheduler** (e.g. `node-cron` or a lightw

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Tier 2 — Close the documented-vs-implemented business-rule gaps

5. **Fee-clearance / attendance-percentage gating for exam eligibility** — the single largest gap between the planning docs and the implemented system (see

11-business-logic-and-sync-matrix.md §11.8). The Workflow Rule Engine's `condition` JSON is the natural mechani

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6. **Attendance correction approval workflow** (maker-checker, 48-hour window) as originally specified — c

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8. **Revaluation workflow** as a distinct, audited process rather than a direct result `PATCH` (§8.11, §11.10).
9. **CGPA/SGPA aggregation** across

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Tier 3 — Scale & robustness

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and `lib/api.ts` (refresh/retry logic), given how much of the security model lives there.

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Tier 4 — Product expansion (per the original planning docs' aspirations)

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18. **Parent/Guardian portal** (unbuilt; planning docs note it can be a read-only role mapped to existing Student IDs).
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15.2 Appendix — Glossary

Term	Meaning in this system
Teaching Assignment	The record linking one faculty member to one…
Curriculum Mapping	The join record placing a subject within a specific…
Result Entry Task	A per-(exam, teaching-assignment) unit of work driving th…
Mentoring Head	A faculty capability flag (`is_mentoring_head`) granting…
HOD	Head of Department — a `faculty.designation` value, DB-…
Registry	An internal, non-routed facade service consolidating a…
Workflow Rule / Workflow Engine	The configurable, event-driven automation system (trigger…
Exam Session vs. Examination Session	Two distinct tables — see [06-database.md](06-…
Academic Session	Not a real table — `students.academic_year` (a string)…
Soft delete	`deleted_at TIMESTAMPTZ`, `NULL` = live row;…
Ownership model	The convention that each database table has exactly one…
RBAC label (frontend)	The richer, client-side-only role vocabulary (`Super Admin…
Simulation Context	A legacy/demo local-state store (`localStorage`-persisted)…

15.3 Appendix — Abbreviations

Abbreviation	Meaning
ERP	Enterprise Resource Planning
RBAC	Role-Based Access Control
JWT	JSON Web Token
LMS	Learning Management System
ATKT	Allowed To Keep Terms (backlog/supplementary exam…
HOD	Head of Department
CGPA/SGPA	Cumulative / Semester Grade Point Average
TTL	Time To Live (cache expiry)
CRUD	Create, Read, Update, Delete
FK / PK	Foreign Key / Primary Key
CSRF / XSS	Cross-Site Request Forgery / Cross-Site Scripting
SAD	Software Architecture Document (this document)

CHAPTER 49
Conclusion

Final evaluation of system design successes and architectural integrity.

In conclusion, the Sreyas College ERP represents a production-grade academic management system utilizing strict architectural layering and reliable data models.

By implementing raw SQL transactions instead of heavy ORMs, the application achieves highly optimized execution speeds and complete control over database constraints.

The separation of concerns between client views, middleware validation filters, service classes, and data schema migrations creates a scalable environment for future institutional features.

CHAPTER 50
References

References, standards, and framework source manuals.

15.4 Appendix — References

- In-repo planning documents: `project_vision_and_requirements.md`, `ARCHITECT_REVIEW.md`, `SYSTEM_MODEL.md`, `BUSI

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- In-repo engineering notes: `docs/erp-integration-audit.md`, `docs/teaching-assignment-refactor-plan.md`.
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15.5 Appendix — Development Notes

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e literal path segments before `/:id` — follow this convention for any new endpoint (see 09-api-documentation.md §9.0).

- Every

new mutating table should get a `deleted_at` column **unless** it is a genuine append-only ledger (attendance-like) — see 06-database.md

d) § 6.5 for the existing precedent either way.

- New cross-module lookups should go through a Registry facade (or a new one should be added) rather than importing a sibling's repository directly.

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15.6 Undocumented or Missing Components

Per the documentation brief, this section explicitly lists everything that could **not** be documented because it does not exist in the codebase, or exists only as a stub/placeholder — rather than silently omitting it.

Entirely missing (no code found)

- Library management module (catalog, checkout/return, fines).
- Hostel management module.
- Transport management module.
- Parent/Guardia

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Hub posting board exists.

- Payment gateway integration (fees are recorded manually only).
- Email/SMS notification delivery (notifications are in-app/Soc

ket.IO only; no SMTP/Twilio-equivalent integration).

- Background job scheduler / cron / message queue of any kind.
- CI/CD pipeline (no `.github/workflo

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- Backend ESLint configuration (backend has no lint config at all, only TypeScript strictness + tests).
- Multi-language / i18n support.
- Biometric hardwar

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- Scheduled purge of expired/revoked `refresh_tokens`.
- Generic department-scoped authorization middleware (HOD department-scoping is handled ad hoc

per-service, not centrally).

- Formal Academic Session entity (`teaching_assignments.academic_session_id` exists as an untyped, unused-by-FK placeholder column).

Partially implemented / stubbed

- **Bulk Academic Data Import**: only the Student entity-type strategy is implemented; Faculty/Subjects/Curriculum/Teaching-Assignment/Department/Clas

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OT_IMPLEMENTED` if invoked (see 05-backend.md §5.15).

- **Timetable conflict detection**: a `POST /timetables/validate-input` endpoin

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in this research pass — verify directly before relying on it (see 11-business-logic-and-sync-matrix.md §11.6).

- **CGPA

/SGPA computation**: no dedicated aggregation function was located; the student dashboard mockups in `UI_UX_BLUEPRINT.md` show a CGPA figure whose cu

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was found — it may require manual intervention today.

- **Fee-based / attendance-based exam eligibility gating**: schema and planning docs describe this; no e

nfor cing cod e pa th w as f oun d in Exa min atio ns/ Res ult s ervi ces (see [11] (11- busi nes s-lo gic- and -syn c-m atrix .md) §11 .8 — the mos t sig nific ant gap flag ged in this

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- **Atten dan ce c orre ctio n ap prov al w orkfl ow**: no c orre ctio n-wi ndo w ti mes tam p or a ppr oval -qu eue tabl e ex ists; corr ecti ons are a dir ect role- gate d `P

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- **Sock et.IO connecti on-ti me auth enti cati on**: a t oke n is pas sed at h and sha ke, but serv er-si de v erifi cati on of that toke n be fore acc epti ng the con nect

ion was not confirmed in the middle ware/server code reviewed — flagged as a security item to verify, not a confirmed vulnerability.

- **Reevaluation & Supplementary** — a student who fails a course may apply for reevaluation of their answer script within 7 days of result publication. A supplementary examination is conducted for students who fail one or two theory papers, typically within 30 days of the main result.
- **Revaluation** — distinct from reevaluation; a fresh valuation of the answer script by a different examiner, permitted only for specific cases and at additional cost.
- **Rubric** — a scoring guide that lists the criteria and levels of performance for a subjective answer, used to make marking consistent across examiners.
- **Scrutiny** — a check that all questions have been marked and totals are correctly added, without re-marking the answers.
- **Semester** — a six-month academic term; Sreyas runs two semesters per year, January–June and July–December.
- **Session** — a named academic year, for example "2025-26", used to group batches and fee structures.
- **Soft delete** — marking a record as deleted by setting a flag and timestamp rather than removing it from the database, so that it can be restored and its history preserved.
- **Supplementary examination** — see reevaluation & supplementary.
- **Transcript** — a consolidated statement of a student's marks and grades across all semesters, issued on request.
- **Valuation** — the act of assigning marks to an answer script by an examiner.
- **Viva voce** — an oral examination, usually conducted for project work or practical courses.

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Superseded but retained (not "missing," but worth flagging as historical)

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ad-only compatibility VIEW over `teaching_assignments`.

- `/api/subject-allocations` routes — retained only as a deprecated, zero-logic re-export of the Teachin

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same student simultaneously (see [11](11-business-logic-and-sync-matrix.md) §11.5).

This list should be treated as a living punch list — re-verify it against the codebase before using it to plan new work, since the system is under active development (most recent commits touch curriculum scheme and teaching assignments, per the repository's git log).